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Effects of Cosmetic and Other Camouflage Interventions on Appearance-related and Psychological Outcomes among Adults with Visible Differences in Appearance: A Systematic Review

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Effects of Cosmetic and Other Camouflage Interventions on Appearance-related and Psychological Outcomes among Adults with Visible Differences in Appearance: A Systematic Review

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ABSTRACT

Objective: Visible differences in appearance are associated with poor social and psychological outcomes. Effectiveness of non-surgical cosmetic and other camouflage interventions is poorly understood. The objective was to evaluate effects of cosmetic and other camouflage interventions on appearance-related outcomes, general psychological outcomes, and adverse effects for adults with visible appearance differences.

Design: Systematic review.

Data Sources: MEDLINE (Ovid), EMBASE (Ovid), PsycInfo (Ovid) CINAHL, and Cochrane Central databases searched from inception through October 24, 2020. Two reviewers independently reviewed titles and abstracts and full-texts.

Eligibility Criteria: Randomised controlled trials in any language on non-surgical cosmetic or other camouflage interventions that reported appearance-related outcomes, general psychological outcomes, or adverse effects for adults with visible appearance differences.

Data Extraction and Synthesis: Two reviewers independently extracted data, assessed intervention reporting using the Template for Intervention Description and Replication (TIDieR) checklist, and assessed risk of bias using the Cochrane risk of bias tool. Outcomes included appearance-related outcomes, general psychological outcomes (e.g., depression, anxiety), and adverse effects.

Results: One head-to-head trial and five trials with waiting list or routine care comparators were included. All had unclear or high risk of bias in at least five of seven domains. Effect sizes could not be determined for most outcomes due to poor reporting. Between-group statistically significant differences were not reported for any appearance-related outcomes and for only five of twenty-five (20%) other psychological outcomes. Given heterogeneity of populations and interventions, poor reporting, and high risk of bias, quantitative synthesis was not possible.

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Conclusions: Conclusions about effectiveness of non-surgical cosmetic or other camouflage interventions could not be drawn. Well-designed and conducted trials are needed. Without such evidence, clinicians or other qualified individuals should engage with patients interested in cosmetic interventions in shared decision making, outlining potential benefits and harms, and the lack of evidence to inform decisions.

Funding Source: Canadian Institutes of Health Research (CMA – 151728).

Registration: PROSPERO (CRD42018103421).

Key Words: Appearance, body image, visible appearance differences, disfigurement, cosmetic, camouflage, systematic review.

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STRENGTHS AND LIMITATIONS OF THIS STUDY

- The systematic review was registered prior to initiation in PROSPERO and reported consistent with standards articulated in the PRISMA statement.
- Review methods aligned with best-practice standards, such as those elaborated in the Cochrane Handbook.
- A peer-reviewed database search strategy was used to ensure rigour and comprehensiveness.
- The Template for Intervention Description and Replication (TIDieR) checklist was used to assess completeness of reporting of intervention characteristics.
- A patient advisory team was involved in reviewing and interpreting results and providing suggestions for implications.

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INTRODUCTION

Visible differences in physical appearance due to medical conditions or injuries are associated with poor social and mental health outcomes.[1-3] Individuals living with visible appearance differences experience unique social challenges, such as staring and invasive questions, and many experience fear of negative evaluation, social interaction anxiety, and social isolation.[1, 4] Intrapersonal consequences include body image dissatisfaction, low self-esteem, and high levels of emotional distress.[1-6]

Interventions that have been used to attempt to improve outcomes among people with visible appearance differences include surgical,[6, 7] psychological,[2, 8-10] and non-surgical cosmetic and other camouflage [11, 12] interventions. Non-surgical cosmetic and other camouflage interventions are comprised of non-invasive techniques intended to minimize or disguise visible appearance differences.[11, 12] These may include the use of cosmetics (e.g., concealer), prosthetics (e.g., prosthetic eye), clothing and stylistic accessories (e.g., scarves), or hair (e.g., wigs).[1]

A 2018 systematic review examined the effects of cosmetic camouflage in children and adults with disfiguring skin disorders on quality of life outcomes.[11] It included cosmetic interventions but not other types of camouflage interventions, and only included people with skin conditions but not appearance differences that were congenital, acquired from medical conditions, treatments, or injuries. Only 6 of 18 included studies were randomised controlled trials (RCTs), and the systematic review did not compare any outcomes between participants randomised to receive camouflage interventions and participants in control arms. Instead, the systematic reported only within-group pre-post mean reductions in Dermatological Life Quality Index scores for the nine studies with pre-post data that reported that measure.

Pre-post analyses of intervention effects, however, are highly prone to bias, and when there is

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no control group it is not possible to determine with any degree of confidence whether an intervention had an effect or to attempt to quantify effects. This would require a precise understanding of the natural history of symptoms or to be able to confidently assume that symptoms do not change over time without intervention. This is typically not the case, however, in trials of interventions that attempt to improve mental health or other areas of subjective well-being. In these areas, participants often seek services and enrol in trials when they are experiencing high levels of symptoms or feeling poorly, and regression to the mean is common.[13-17] In major depression intervention trials, for instance, approximately 40% of participants assigned to placebo groups in drug trials or no-treatment groups in psychological intervention trials achieve remission, which has been termed "spontaneous recovery".[18] The Cochrane Handbook discourages inclusion of evidence from non-randomised studies when conducting trials is highly feasible and when evidence from non-randomised trials would be unlikely to provide interpretable results.[19]

We conducted a systematic review with the aim of informing health care providers and the public about the effectiveness of cosmetic and other camouflage interventions to improve outcomes among people with visible appearance differences or, in the absence of informative evidence, to identify this shortcoming and describe improvements that are needed. Thus, the objective was to evaluate evidence from RCTs of cosmetic and other camouflage interventions with training or instructions delivered (a) in person or (b) as self-help material (e.g., leaflets, internet-based) on appearance-related outcomes, general psychological outcomes, and adverse effects among adults with appearance differences that were congenital, from injury, a medical condition, or medical treatment.

METHODS

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This systematic review was registered in PROSPERO (CRD42018103421), and results were reported per the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement.[20]

Identification of Eligible Studies

Eligible studies were reports of RCTs in any language that examined the effects of non-surgical cosmetic or other camouflage interventions on appearance-related or general psychological outcomes among adults (ages 18 and over) with congenital or acquired visible appearance differences from injury (e.g., burn injury), disease or medical condition (e.g., scleroderma), or medical treatment (e.g., resection of facial tumor, breast reconstruction). Studies that included both children and adults were only included if results for the adults were reported separately or if greater than 80% of participants were adults. Trials that enrolled only individuals with visible appearance differences due to self-harm were excluded. Eligible interventions were defined as interventions with a core component that involved teaching, modelling, or demonstrating the modification or enhancement of some aspect of physical appearance through non-surgical cosmetic or other camouflage strategies. Eligible intervention formats included individual and group formats and delivery via any mode (e.g., in-person, web-based, bibliotherapy) by any type of provider. Eligible comparators included (1) any inactive control condition (e.g., usual care, attention control, sham procedure); (2) a non-cosmetic intervention designed to target appearance concerns (e.g., psychological intervention); and (3) another eligible cosmetic intervention. To be eligible for inclusion, studies must have reported (1) appearance-related psychological outcomes; (2) general psychological outcomes, including symptoms of depression or anxiety; or (3) adverse effects from the intervention.

Search Strategy

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A health sciences librarian (JB) developed the search strategy and performed the literature searches in MEDLINE (Ovid), EMBASE (Ovid), PsycInfo (Ovid) CINAHL, and Cochrane Central from database inception until October 24, 2020 with no language restrictions. The Cochrane Highly Sensitive Search Strategy for identifying randomised trials in MEDLINE: sensitivity-maximizing version (2008 revision) was used.[21] The MEDLINE strategy was developed with input from the project team and peer-reviewed by a second librarian using the Peer Review of Electronic Search Strategies (PRESS) guidelines.[22] After the initial MEDLINE strategy was finalized, it was adapted for use in the other databases. The complete search strategy is available in Supplementary File 1. We also reviewed reference lists of included articles and relevant reviews and conducted a forward citation search via Google Scholar.[23] We searched ClinicalTrials.gov (clinicaltrials.gov), ISRCTN (www.isrctn.com), and the World Health Organization registry search portal (apps.who.int/trialsearch) to identify any additional trials that may have been initiated but not published.

Search results were initially downloaded into the citation management database RefWorks (RefWorks, RefWorks-COS, Bethesda, MD, USA) for duplicate removal. Unique citation records were then transferred into the systematic review program DistillerSR (Evidence Partners, Ottawa, Canada). DistillerSR was used to track results of the review process. Two investigators independently reviewed titles and abstracts for eligibility. If either reviewer deemed a citation potentially eligible based on a review of the title and abstract, a full-text review of the article was completed by two reviewers independently. Any disagreement between reviewers after full-text review was resolved by consensus, including consultation with an independent third reviewer if necessary.

Data Extraction and Analysis

Data were extracted from included RCTs by two investigators using a standardized electronic data collection form in DistillerSR.

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Disagreements were resolved by consensus and, if necessary, consultation with an independent third reviewer. See Supplementary File 2 for a complete list of extracted variables.

We used the Template for Intervention Description and Replication (TIDieR) checklist [24] to evaluate the degree to which interventions were reported sufficiently to allow replication in research or practice. The checklist is comprised of 12 items that assess reporting of the intervention name; the rationale or theory underlying the intervention; physical or informational material used; procedures and processes of the intervention; the intervention provider and their background; the mode of delivery, such as group or face-to-face; the location where the intervention was delivered and necessary infrastructure; the number of sessions, schedule, and duration; if any tailoring was done and how; any modifications made to the intervention; if adherence or fidelity was assessed and how; and, if adherence or fidelity was assessed, the extent to which the intervention was delivered as planned.

We used the 2011 version of the Cochrane Collaboration's risk of bias tool to evaluate risk of bias.[25] The tool has 7 domains, including random sequence generation, allocation concealment, blinding of participants and personnel, blinding of outcome assessment, incomplete outcome data, selective outcome reporting, and any other potential sources of bias. Studies were rated low, unclear, or high risk of bias on each domain.

We did not pool results across trials because of the heterogeneity of interventions and trial designs and concerns about risk of bias. Rather, we descriptively reported trial characteristics and outcomes. Between groups standardized mean differences (SMDs) were calculated for each relevant outcome using Hedges's *g* [26] when possible. For trials where sufficient data were not provided to

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calculate SMDs, other relevant information that could be used to describe intervention outcomes (e.g., median change in each trial arm with p value) was reported.

Patient and Public Involvement

We assembled a Body Image Patient Advisory Team comprised of four female and one male persons with scleroderma, a rare disease that often causes changes in appearance, to advise on the present systematic review. The advisory team participated in a patient panel to discuss the study objectives, findings, and implications of the study. Members of the patient panel were also sent copies of the present manuscript prior to submission and asked to provide any feedback or suggestions. Members of the advisory team commented on issues related to characteristics of participants in the trials we reviewed, the acceptability of camouflage interventions, and delivery of these interventions via health care providers. These comments were incorporated into the Discussion section of the manuscript.

Modifications to Protocol

When we developed the review protocol, we did not anticipate that we might encounter trials that enrolled only individuals with visible appearance differences from self-inflicted injuries, and eligibility criteria did not address this group. We subsequently added visible appearance differences from self-injury as an exclusion criterion because we deemed the population to be substantively different than the target population; interventions for this population would need to address factors related to self-harming behaviour, in addition to appearance, which would be out of scope for our intended target population.

RESULTS

Search Results

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The combined database searches identified 1,085 unique titles and abstracts. Of these, 1,062 were excluded after screening titles and abstracts and 19 after full-text review, resulting in 4 eligible RCTs. One additional eligible RCT [27] was identified via a database search that we conducted for a separate systematic review on psychological interventions (PROSPERO: CRD42018100632), and a second additional eligible RCT [28] was identified via manual searching of a relevant systematic review,[11] resulting in 6 included RCTs.[27-32] See Figure 1. No additional eligible trials were identified via manual searching of reference lists of included articles, citation tracking, or review of trial registries. One ongoing trial was identified via the review of trial registries (NCT03540966), but results were not available as of November 2, 2020.

Characteristics of Included Studies

The 6 included RCTs were published between 2006 and 2019.[27-32] Five trials compared an intervention to a no-intervention, waitlist, or usual care control,[27, 29-32] and one trial [28] was a head-to-head comparison of two different camouflage products.

Comparison to No Intervention or Usual Care

Trial participants included survivors of gynaecologic malignancies with chemotherapy-induced alopecia,[27] head and neck cancer survivors,[29] people living with acne vulgaris,[30] head and neck skin cancer survivors,[31] and survivors of breast cancer.[32] The number of participants per trial ranged from 24 to 136 (mean: 65; median: 50), and there was a total of 324 participants, with 162 participants each in intervention and control arms of trials. All participants were female. The mean ages for participants in each trial ranged from 29 to 67. See Table 1.

The trials were conducted in the United States,[27] Taiwan,[29] Japan,[30] Italy,[31] and

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Germany.[32] The interventions ranged from one to four sessions. Formats included videotape review,[27], combined individual and group sessions,[29] combined individual sessions and videotape,[30] and group sessions. [31, 32] Two RCTs required either self-report or investigator confirmation of appearance changes from an illness, injury, or medical procedure as a condition of eligibility;[29, 32] none, though, set a threshold for severity. No trials included patient report of concern about appearance as part of eligibility criteria, even though the trials targeted appearance-related and psychological variables.

In two RCTs, participants were provided with cosmetic camouflage products.[30, 31] In one trial,[30] which included people with acne, all products were developed and provided by Nov Pharmaceuticals and included skincare and cosmetic products and instructions on using the products that were specific to people with acne. In the second trial, customized camouflage plans that were first submitted to participants for approval were provided, and cosmetic camouflage products were provided by Istituto Ganassini Corporate;[31] however the specific cosmetic products were not described. Three of the trials [27, 29, 32] provided instructions on using camouflage techniques but did not provide specific products. In one trial,[27] participants viewed a 45-minute video that provided camouflage suggestions and techniques, including information on cosmetic camouflage and wigs. In another, hands-on instruction and practice with cosmetic camouflage was initiated, and a 25-page manual was provided.[29] In another, a group makeup workshop was provided and followed with a professional photoshoot and delivery of edited photographs.[32]

Comparators included standard counselling on hair loss,[27] unspecified routine care,[29, 31] conventional acne treatment,[30] and a wait-list control.[32] Four trials included appearance-related outcome measures,[27, 29, 31, 32] two trials included a depression symptom outcome measure,[29, 32] one trial included an anxiety outcome

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measure,[29] and five trials included general mental health outcome measures.[27, 29-32] No trials reported on adverse effects from participation.

Head-to-head Trial

The head-to-head trial,[28] which was conducted in France, included 88 healthy female volunteers with various facial aesthetic imperfections (e.g., melisma, scars from chicken pox), the presence of which were validated by clinical examination. The mean age of participants was 30. Participants were randomised to receive either a heavier-coverage liquid or a lighter-coverage powder product; all participants received recommendations and instructions from a professional makeup trainer in a single individual session. The study outcome was a body image-specific quality of life measure. See Table 1.

Synthesis of Outcomes

Comparison to No Intervention or Routine Care Controlled Trials

There was sufficient information to calculate intervention effect sizes (SMDs) and confidence intervals for two of the five trials that compared an intervention group to a no intervention, wait list, or routine care control group,[29, 30] only one of which [29] included an appearance-related outcome.

Appearance-related Outcomes: In the four trials that reported appearance-related outcomes,[27, 29, 31, 32] there were no statistically significant differences between camouflage intervention and control groups. Reported effect sizes were generally negligible. See Table 2.

General Psychological Symptoms: Across a total of 25 general psychological outcome measurement points reported in the five trials, five (20%) were statistically significant. There was a small to moderate effect on the Leibowitz Social Anxiety Scale-Anxiety subscale at 8 weeks (SMD: -0.41 CI: -0.89, -0.07) in a trial that randomised 70 head and neck cancer survivors to individual and

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group sessions or routine care,[29] although the effect was smaller and not statistically significant at 4 or 12 weeks, and the effect for the Fear subscale was non-significant at all three time points. There were moderate to large magnitude effects for depressive symptoms (*d*: -0.83 at 2 weeks and *d*: -0.43 at 4 weeks) and self-esteem (*d*: 0.68 at 2 weeks and *d*: 0.72 at 4 weeks) in a trial that randomised 44 breast cancer patients to a single group session versus wait-list control, though it was not clear if they were statistically significant.[32] See Table 2.

Head-to-head Trial

In the head-to-head trial, which compared liquid and powder camouflage products among 88 people with a variety of aesthetic imperfections,[28] sufficient information to calculate effect sizes was not provided. Of the four outcomes measured at two time points (8 outcomes total), one was reported to be statistically significantly different between groups (a self-confidence subscale at 1 week).

Reporting of Intervention Components and Risk of Bias

As assessed using the TIDieR checklist, few trial reports described the interventions used with sufficient detail to fully understand what had occurred or to consider replication (See Table 3). Materials were not sufficiently reported in terms of describing specific materials used or providing details on how to access the materials for all of the trials.[27-32] Although all of the studies described who provided the intervention, four did not provide details about training or qualifications of the interventionists specific to camouflage or visible appearance differences.[28, 30-32] Adherence was not assessed in any of the six trials.

Risk of bias was evaluated using the Cochrane criteria with most studies rated as high or unclear across the domains assessed (See Table 4). Random sequence generation procedures were only described in two of the studies.[29, 31] Due to the nature of the interventions and self-reported

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psychological outcomes used in the studies, risk of bias due to a lack of blinding of participants/personnel and outcome assessment was high across all studies. Overall, of the seven domains assessed (including "other"), three RCTs were rated as unclear or high in five domains,[27, 29, 32] two in six domains,[30, 31] and one in all seven domains.[28]

DISCUSSION

We identified five RCTs that compared camouflage interventions to no treatment, waiting list, or routine care controls, and one head-to head trial that compared two camouflage products. Trials included participants from populations with a range of visible appearance differences, including people with head and neck cancer, acne vulgaris, facial skin cancer, chemotherapy-induced alopecia, breast cancer, and non-defined facial aesthetic imperfections. Risk of bias was unclear or high in at least five of seven possible domains for all included RCTs. Furthermore, outcomes were incompletely reported in many trials, and we could only calculate effect sizes for a small proportion of outcomes that were described. Very few statistically significant results were identified. Additionally, based on TIDieR checklist ratings, interventions were generally not described with enough information to support replication or use in practice. Given these limitations, a meaningful quantitative synthesis of the findings from included RCTs was not possible, and conclusions about the likely effectiveness of non-surgical cosmetic interventions on appearance-related or general psychological outcomes could not be drawn with any degree of confidence. No trials reported on adverse effects.

A previous review [11] used pre- and post-intervention scores among only studies that used a single quality of life measure, the Dermatology Life Quality Index (DLQI),[33] and concluded that reductions in scores across nine studies of varying designs suggested effectiveness of cosmetic camouflage on quality of life outcomes. That review did not attempt to synthesize between-group effects or describe any limitations due to not evaluating between-group effects. There are several

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reasons that help explain why our conclusion differs from the conclusion of that review. First, the specific focus of the previous review was the role of cosmetic camouflage on quality of life outcomes, whereas the present review was broader in scope and included any non-surgical cosmetic intervention (e.g., wigs, prosthetics) and any appearance-related and general psychological outcomes. Second, the previous review included studies of children and adults, whereas our review was limited to adult samples. Third, and most importantly, the conclusions of the previous review were drawn from pre-post comparisons without control groups, whereas we included only RCTs. RCTs are considered the gold standard study design for establishing the effectiveness of psychological interventions and inferring causation.[34] There is a high risk in trials of psychosocial interventions, in which people often consent to participate when they are struggling with concerns, that the natural trajectory of included participants may be to improve over time. Non-randomised trials in this area are pre-post trials that must assume that there would be no change over time without the intervention. Systematic reviews that draw conclusions and make recommendations from these kinds of trials are at high risk of misleading clinicians who seek to provide the best possible care to people with visible differences in appearance, misleading people with visible appearance differences about their options, and losing the opportunity to encourage researchers to engage in improved trials that answer a critical health care question.

Several important methodological limitations were noted in the RCTs included in the present review, and there are implications for future research. First, study aims were often ambiguous, and it was unclear if what was being tested was a specific product, camouflage in general, or instructions on using camouflage. Second, few trials provided information that would be needed for replication or use in practice. Third, no studies had eligibility criteria for baseline appearance concerns or subjective severity. Only three of the trials [28, 29, 32] included eligibility criteria relating to confirming the

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presence of appearance differences; however, no trials included a threshold for objective severity or required that participants had appearance-related concerns. Thus, some participants may have been enrolled even though they did not have appearance concerns, and those participants would not have been able to benefit from the interventions. Fourth, none of the studies reported on harms from the interventions. It is possible that some products may exacerbate skin conditions and lead to worse objective appearance or psychological outcomes. Fifth, no studies included men, limiting generalizability of the findings. This was a limitation that was emphasized by the Patient Advisory Team, which raised the concern that the acceptability of camouflage among men is not known. Sixth, the small number of RCTs available, the generally small sample sizes included, and the poor reporting of trial methods and results limited utility of the trials and the ability to draw conclusions from what they reported; these are common issues in rare disease and visible appearance differences research.[35]

There are clinical implications of this review. Because of high patient interest in non-surgical cosmetic treatments [12] and the fact that for some conditions causing visible appearance differences non-surgical cosmetic treatments are the only treatment option available, many people may consider using camouflage or other cosmetic interventions. A concern raised by members of our Body Image Patient Advisory Team, however, was that it may be better to accept one's visible appearance differences rather than engaging in behaviours to attempt to hide or conceal them. It was noted that suggesting camouflage may send an undesired message to patients who are comfortable with their appearance or who are working to be more comfortable. In addition to improving the quality and reporting of trials, future researchers may consider the role of internet delivery of interventions, as health interventions are increasingly delivered via the internet and have been shown to be similarly

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effective to face-to-face interventions.[36-38] Online delivery could enable wider recruiting and larger sample sizes as well as more efficient dissemination if an intervention is found to be effective.

Patients who are interested in these approaches and may wish to discuss advantages and disadvantages with health care providers or other qualified individuals. Given the lack of high-quality evidence, clinicians are left with a gap not only around whether interventions are likely to be effective, but whether there may be adverse effects to consider. In light of these limitations, we recommend that clinicians working with visible appearance differences populations engage in a shared decision making approach with patients who express an interest in or who may benefit from non-surgical cosmetic intervention[39], such that education is provided on the current lack of empirical information about clinical and psychological harms as well as effectiveness of interventions.

The findings of this review should be interpreted in the context of several limitations of the review itself. First, although a systematic search designed by a medical librarian was utilized, incompleteness of the evidence reported and publication bias limiting null results are concerns for any systematic review. Second, the small number of studies eligible for inclusion in the review and differences in how data were reported did not allow for publication bias to be assessed. Third, given the low quality of trial evidence available, in addition to not being able to draw conclusions about overall effectiveness, examination of potentially relevant participant (e.g., age, sex, marital status) and condition (e.g., years of living with the visible appearance difference, patient-perceived and observer-rated severity) variables could not be undertaken. Fourth, even with better quality evidence, the inconsistency across studies in outcome domains and measures used to evaluate effects would make drawing conclusions difficult. Ideally, a core set of outcome domains and measures would be used consistently.

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In summary, the present review did not find evidence to evaluate benefits or harms of the use of non-surgical cosmetic treatments in visible appearance differences. There are several recommendations for future RCTs evaluating the efficacy of non-surgical cosmetic interventions in visible appearance differences populations, such as the inclusion of appearance-related eligibility criteria and the use of internet-based and other feasible delivery modalities. Until more robust scientific evidence becomes available, clinicians can engage in shared decision making with patients to determine whether engaging in non-surgical cosmetic interventions given the current lack of empirical support may be warranted.

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AUTHORS' CONTRIBUTIONS

SG, DBR, ACJ, and BDT were responsible for the study concept and design, drafted the study protocol, contributed to data extraction, contributed to drafting the manuscript, and approved the final manuscript. JB designed and conducted the search strategies, contributed to drafting the manuscript, and approved the final manuscript. VLM and LK were responsible for the study concept and design, drafted the study protocol, contributed to drafting the manuscript, and approved the final manuscript. BDT is the guarantor.

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COMPETING INTERESTS STATEMENT

All authors have completed the ICMJE uniform disclosure form at www.icmje.org/coi_disclosure.pdf and declare that they have no competing interests.

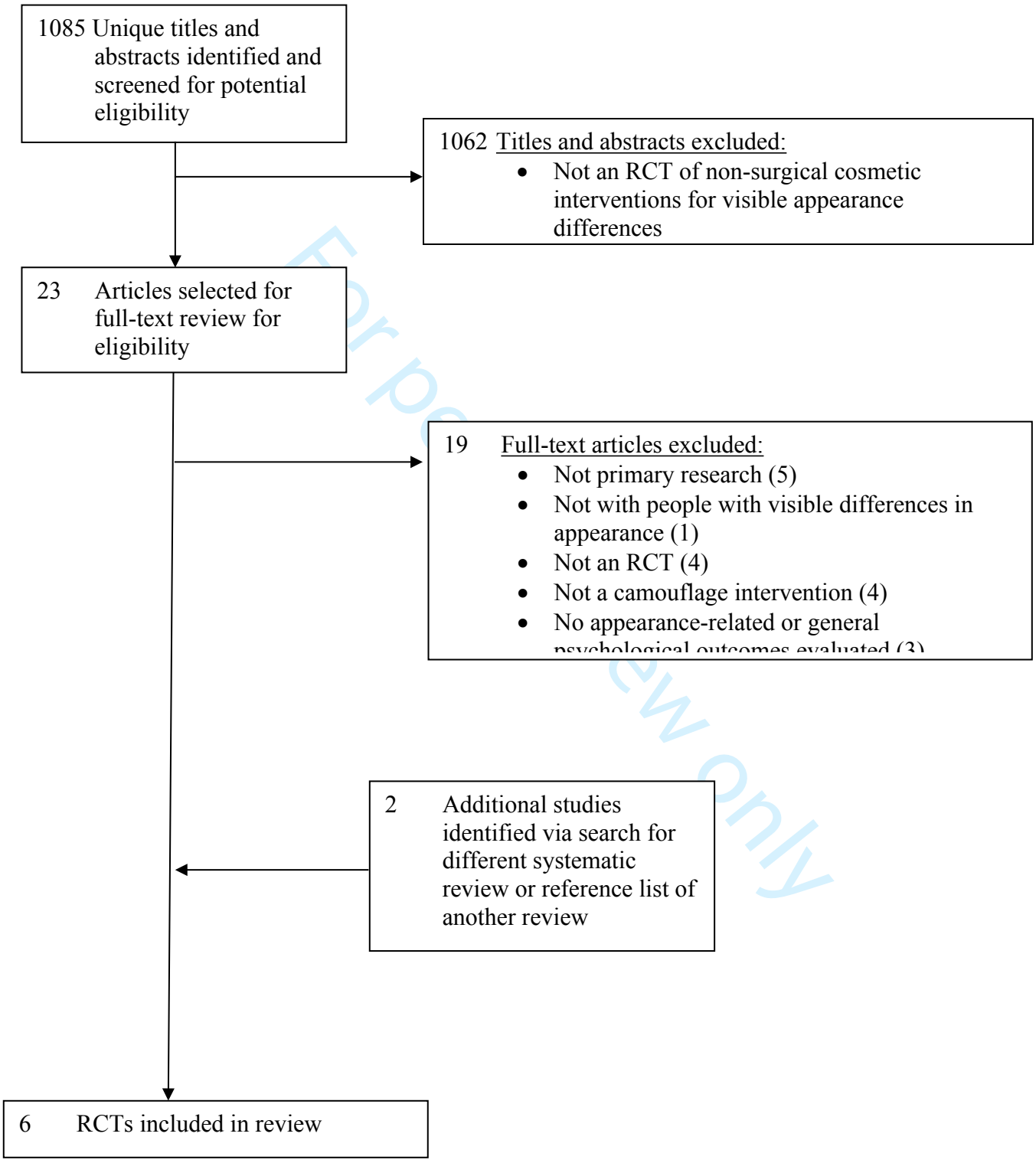
DATA SHARING STATEMENT

All data extracted and analysed for the present study are reported in the manuscript and its tables.

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Figure 1. PRISMA flow diagram of study selection



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Table 1. Characteristics of included studies

Author Year	N Randomised	Country	Population	Age mean (sd)	Format	Materials Provided to Participants	Number of Sessions	Length of Session(s)	Comparator
No Intervention or Routine Care Controlled Trials									
Chen 2016 [29]	Total: 70 Tx: 35 Control: 35	Taiwan	Head and neck cancer survivors	52 (2)	Individual and group: Led by research nurses, a research cosmetologist, and cosmetology assistants	Educational manual	4 sessions (3 individual; 1 group)	Up to 3 hours	Routine care
Matsuoka 2006 [30]	Total: 50[a] Tx: 25 Control: 25	Japan	People with acne vulgaris	25 (4)	Individual and videotape: Led by patient's primary dermatologist with taped instructions	Unspecified camouflage products	1 session	Not reported	Standard acne treatment
Nicoletti 2014 [31]	Total: 24 Tx: 12 Control: 12	Italy	People with facial skin cancer	67 (NR)	Group: Led by professional makeup artist; psychologist also in attendance	Unspecified camouflage products	3 sessions	2 hours	Routine care
Nolte 2006 [27]	Total: 136 Tx: 68 Control: 68	United States	People with chemotherapy-induced alopecia	58 (12)[b]	Videotape: cosmetic camouflage and wig/hair covering techniques	No products provided	1 session	45 minutes	Standard counselling on hair loss
Richard 2018 [32]	Total: 44 Tx: 22 Control: 22	Germany	People with breast cancer	39 (9)	Group: makeup workshop, photoshoot, and receipt of edited photos	Unspecified camouflage products used in session but not provided; edited photographs	1 session	4 hours	Wait-list control
Head-to-Head Trial									
Beresniak 2017 [28]	Total: 88[c]	France	People with facial aesthetic imperfections	30 (8)	Individual: Recommendations and instructions provided by professional makeup trainer	Liquid or powder camouflage products	1 session	Not reported	Other camouflage product

[a]Number randomised not provided, authors noted that 50 patients participated in the study; [b]This is the age for N = 187 women in broader sample from which N=136 were drawn; [c]Number per group not provided; authors noted that 88 patients were randomised.

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Table 2. Outcomes of Included Trials. Scores reversed for some outcome measures such that higher scores indicate worse functioning for all measures and negative between-group effects reflect better outcomes in the intervention groups.

Study	N randomised	Appearance-related	General Mental Health	Depression	Social Anxiety
No Intervention or Routine Care Controlled Trials					
Chen 2016 [29]	Total: 70[a] Tx: 35 Control: 35	BIS: SMD (95% CI) 4 weeks -0.07 (-0.54, 0.40) 8 weeks -0.08 (-0.56, 0.39) 12 weeks -0.43 (-0.92, 0.06)	RSES: SMD (95% CI) 4 weeks -0.19 (-0.67, 0.28) 8 weeks 0.31 (-0.17, 0.79) 12 weeks -0.03 (-0.52, 0.45)	HADS-D: SMD (95% CI) 4 weeks 0.33 (-0.15, 0.80) 8 weeks -0.08 (-0.56, 0.39) 12 weeks -0.14 (-0.62, 0.35)	LSAS-F: SMD (95% CI) 4 weeks -0.29 (-0.76, 0.19) 8 weeks -0.29 (-0.77, 0.19) 12 weeks -0.41 (-0.89, 0.08) LSAS-A: SMD (95% CI) 4 weeks -0.32 (-0.80, 0.15) 8 weeks -0.41 (-0.89, -0.07) 12 weeks -0.33 (-0.82, 0.16)
Matsuoka 2006 [30]	Total: 50[b] Treatment: 25 Control: 25	NA	WHOQOL-26-P: SMD (95% CI) 4 weeks 0.05 (-0.50, 0.60) WHOQOL-26-SR: SMD (95% CI) 4 weeks -0.02 (-0.57, 0.54) DLQI-SF: SMD (95% CI) 4 weeks 0.00 (-0.55, 0.55) DLQI-PR: SMD (95% CI) 4 weeks	NA	NA

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Study	N randomised	Appearance-related	General Mental Health	Depression	Social Anxiety
			0.16 (-0.40, 0.71)		
Nicoletti 2014[c] [31]	Total: 24[d] Treatment: 12 Control: 12	BUT-GSI: 3 months[e]: *Tx:* change = -0.26, IQR = -0.72, 0.16 *Control:* change = 0.03, IQR = -0.08, 0.34 p = 0.371 6 months[e]: *Tx:* change = -0.15, IQR = -0.54, 0.45 *Control:* change = 0.12, IQR = -0.09, 0.64 p = 0.318	SP-PF: 3 months[e]: *Tx:* change = -4.75, IQR = -12.47, 5.02, *Control:* change = 7.15, IQR = 4.4-8.1 p = 0.270 6 months[e]: *Tx* change = -9.15, IQR = -10.3, -3.25 *Control* change = 18.1, IQR = -4.5, 28.33 p = 0.003 SP-SF: 3 months[e]: *Tx:* change = -3.54, IQR = -11.04, 17.02 *Control:* change = 13.84, IQR = -3.75, 17.92, p = 0.318 6 months[e]: *Tx group* change = -1.99, IQR = -16.00, 9.54 *Control group* change = 0.83, IQR = -5.17, 5.50 p = 0.636	NA	NA
Nolte 2006[h] [27]	Total: 136 Treatment: 68 Control: 68[i]	BCSCS-BC: Post-tx: F[1,132] = 0.11, p = 0.74 (Tx: 68; Control 68)	BCSCS-SC: Post-tx F[1,132] = 0.32, p = 0.57 (Tx: 68; Control 68)	NA	NA

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Study	N randomised	Appearance-related	General Mental Health	Depression	Social Anxiety
Richard 2018 [32]	Total: 44 Treatment: 22 Control: 22[f]	BIS: 2 weeks, 4 weeks: NR; Improvements in body image were independent of treatment (p > 0.05)	RSES: 2 weeks: *d* = -0.68[g] 4 weeks: *d* = -0.72[g]	CES-D: 2 weeks: *d* = -0.83[g] 4 weeks: *d* = -0.45[g]	NA
Head-to-head trial					
Beresniak 2017 [28]	Total: 88[j]	BeautyQOL-Attractiveness: at 1 and 3 weeks, mean A > mean B; p>0.05 BeautyQOL-Social Life: at 1 and 3 weeks, mean A > mean B; p>0.05 BeautyQOL-Mood: at 1 and 3 weeks, mean A > mean B; p >0.05 BeautyQOL-Self Confidence: 1 week: mean A > mean B; p <0.05 3 weeks: mean A > mean B; p >0.05	NA	NA	NA

[a]Tx (*n*: 35, 34, 32) and control (*n*: 34, 34, 34) at 4, 8, and 12 weeks, respectively; [b]Number randomised not provided, noted that 50 patients participated in the study; [c]Effect sizes could not be calculated; [d]Tx (*n*: 12, 12) and control (*n*: 12, 12) at 3 and 6 months, respectively; [e]Data reported as change score (IQR) from baseline to 3 months and p-value is for between-group difference; [f]Tx (*n*: 20, 29) and control (*n*: 19, 19) at 2 and 4 weeks; [g]Baseline corrected; [h]Intervention effect comparing baseline scores to scores after course 4 of chemotherapy using BCSCS scores during course 3 of chemotherapy as a covariate reported; [i]Tx (*n*: 68) and control (*n*: 68) at post-tx measurement; [j]Tx and Control *n* not provided; 86 participants completed study

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Abbreviations: BCSCS-BC (Body Cathexis/Self-Cathexis Scale: Body Cathexis); BCSCS-SC: Body Cathexis/Self-Cathexis Scale: Self-Cathexis; BIS: Body Image Scale (higher scores indicate poorer body image); BIS: Body Image Scale (higher scores indicate poorer body image); BUT-GSI: Body Uneasiness Test Global Score Index (higher scores indicate greater body uneasiness and negative change score indicates improvement); DLQI-PR (Dermatology Life Quality Index: Personal Relationships); DLQI-SF (Dermatology Life Quality Index: Symptoms/Feelings); HADS-D: Hospital Anxiety and Depression Scale: Depression subscale (higher scores indicate greater depression); LSAS-A: Liebowitz Social Anxiety Scale: Anxiety subscale (higher scores indicate greater anxiety); LSAS-F: Liebowitz Social Anxiety Scale: Fear subscale (higher scores indicate greater fear); RSES: Rosenberg Self-Esteem Scale); SP-PF: Satisfaction Profile: Psychological Functionality; SP-SF: Satisfaction Profile: Social Functionality; Tx = Treatment/intervention group; WHOQOL-26-P: World Health Organization Quality of Life: Psychological domain; WHOQOL-26-SR: World Health Organization Quality of Life: Social Relationships domain

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Table 3. TIDieR Checklist Results

Studies	Name	Why	Materials	Procedures	Who provided	How	Where	When and how much	Tailoring	Modifications	How well: Planned adherence	How well: Actual adherence
No Intervention or Routine Care Controlled Trials												
Chen 2016[29]	Yes	Yes	Partial[d]	Yes	Yes	Yes	No	Partial[e]	N/A	N/A	N/A	N/A
Matsuoka 2006[30]	Yes	Yes	Partial[d]	Yes	Partial[b]	Yes	Partial[f]	No	Yes	N/A	N/A	N/A
Nicoletti 2014[31]	Yes	Yes	Partial[g]	No	Partial[b]	Yes	Yes	Yes	Partial[c]	N/A	N/A	N/A
Nolte 2006[27]	Yes	Partial[h]	Partial[d]	Yes	Yes	Partial[i]	Partial[f]	Partial[j]	N/A	N/A	N/A	N/A
Richard 2018[32]	Yes	Yes	Partial[a]	Yes	Partial[b]	Partial[k]	Partial[f]	Yes	N/A	N/A	N/A	N/A
Head-to-head trial												
Beresniak 2017 [28]	Yes	Yes	Partial[a]	Yes	Partial[b]	No	No	Yes	Partial[c]	N/A	N/A	N/A
Total	6 Yes 0 Partial 0 No 0 N/A	5 Yes 1 Partial 0 No 0 N/A	0 Yes 6 Partial 0 No 0 N/A	5 Yes 0 Partial 1 No 0 N/A	2 Yes 4 Partial 0 No 0 N/A	3 Yes 2 Partial 1 No 0 N/A	1 Yes 3 Partial 2 No 0 N/A	3 Yes 2 Partial 1 No 0 N/A	1 Yes 2 Partial 0 No 3 N/A	0 Yes 0 Partial 0 No 6 N/A	0 Yes 0 Partial 0 No 6 N/A	0 Yes 0 Partial 0 No 6 N/A

Note: Yes (Y), No (N), Partial (P), Not applicable (N/A); [a]Specific brand information not provided; [b]Training information not provided; [c]Details regarding how adaptations were made not provided; [d]Information on how to access materials not provided; [e]Duration of all sessions not provided; [f]Details about infrastructure and other relevant feature not provided; [g]Specific products not described; [h]details about rationale underlying intervention not provided; [i]unclear if entire sessions was individual format; [j]Timeline of intervention provision unclear; [k]Number of participants per group/number of groups not specified .

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Table 4. Quality assessment for included RCTs

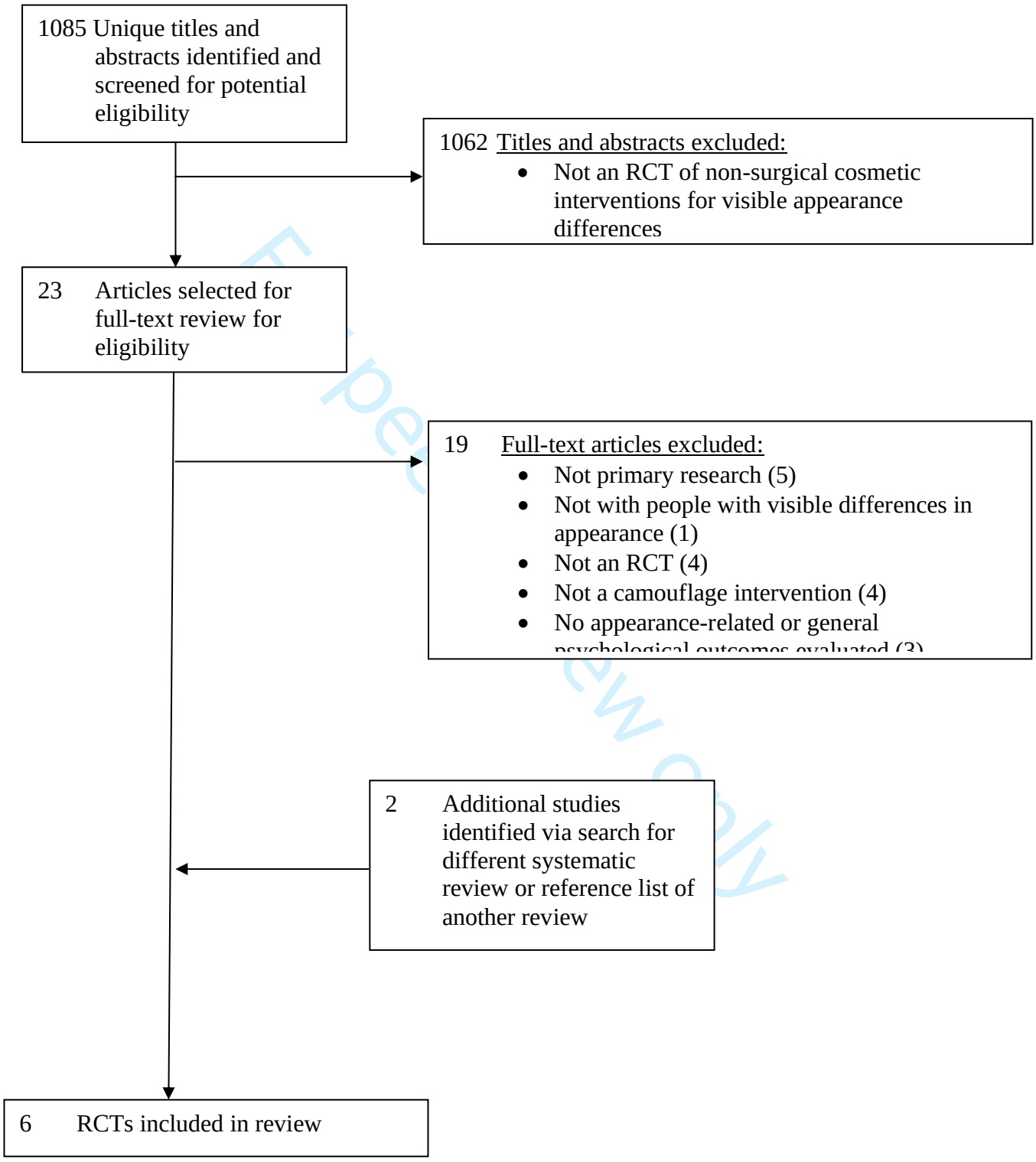
Studies	Random sequence generation	Allocation concealment	Blinding of participants/ personnel	Blinding outcome assessment	Incomplete outcome data	Selective reporting	Free of other bias
No Intervention or Routine Care Controlled Trials							
Chen 2016[29]	Low	Unclear	High	High	Low	Unclear	Unclear[b]
Matsuoka 2006[30]	Unclear	Unclear	High	High	Low	Unclear	Unclear[c]
Nicoletti 2014[31]	Low	Unclear	High	High	Unclear	Unclear	Unclear[c]
Nolte 2006[27]	Unclear	Unclear	High	High	Low	Unclear	Low
Richard 2018[32]	Unclear	Unclear	High	High	Low	Unclear	Low
Head-to-head trial							
Beresniak 2017 [28]	Unclear	Unclear	High	High	High	Unclear	Unclear[a]
Total	2 Low 4 Unclear 0 High	0 Low 6 Unclear 0 High	0 Low 0 Unclear 6 High	0 Low 0 Unclear 6 High	4 Low 1 Unclear 1 High	0 Low 6 Unclear 0 High	2 Low 4 Unclear 0 High

Note. Green indicates low risk of bias; yellow indicates unclear risk of bias; red indicates high risk of bias [a]Only mean scores reported, unclear if product brand and consultant affiliation are related; [b]Significant difference at baseline for experimental group in radiotherapy dosage[c] not controlled for in analysis; [c]All products provided by a single pharmaceutical company.

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Figure 1. PRISMA flow diagram of study selection



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Appendix 1. Search Strategies

Ovid MEDLINE(R) and Epub Ahead of Print, In-Process & Other Non-Indexed Citations, Ovid MEDLINE(R) Daily 1946 to Present

1 exp Cicatrix/
2 exp Facial Injuries/ or exp Craniofacial Abnormalities/ or exp Facial Dermatoses/
3 Psoriasis/ or Eczema/ or Acne Vulgaris/
4 exp Skin Abnormalities/
5 exp Hemangioma/
6 exp Pigmentation Disorders/
7 exp Melanoma/
8 Burns, Chemical/ or Burns/ or Burns, Inhalation/ or Burns, Electric/
9 exp Alopecia/
10 exp Exophthalmos/
11 exp Strabismus/
12 exp Eyelid Diseases/
13 exp Mouth Neoplasms/
14 exp "head and neck neoplasms"/
15 Growth Disorders/
16 exp Dwarfism/
17 Gigantism/
18 exp Dystonia/
19 Dupuytren Contracture/
20 Amputation/
21 Artificial Limbs/
22 Amputees/
23 exp reconstructive surgical procedures/
24 exp Mastectomy/
25 Colostomy/
26 exp Mouth Abnormalities/
27 ((visible or face or facial* or maxillofacial or jaw) adj3 (injur* or damage* or scars or scarring or scarred or dermatos*)).tw,kf.
28 (keloid* or cicatrix or Acne or Eczema or psoriasis or Epidermolysis Bullosa or port wine stain* or h?emangioma$ or vitiligo or birth mark* or birthmark*).tw,kf.
29 (melanoma* or burns or burn or burned or alopecia or exophthalm* or graves disease or thyroid eye disease or strabismus or (misalign* adj3 eye*) or blepharoptosis or ectropion).tw,kf.

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30 ((eye or eyelid or head or neck or mouth or oral) adj3 (neoplasm* or cancer* or tumo?r*)).tw,kf.

31 (Dwarfism or dwarf or small stature or gigantism or restricted growth or torticollis or dystonia or dupuytren* contracture*).tw,kf.

32 (Loss of limb* or amputee* or artificial limb*).tw,kf.

33 (appearance adj3 (abnormal or malformation or problem* or distress)).tw,kf.

34 (visibl* adj3 (differen* or disabilit*)).tw,kf.

35 (deformit* or deformed or disfigured or disfiguring or disfigurement).tw,kf.

36 (palatoschisis or hare lip* or harelip* or (cleft* adj (lip* or palate* or orofacial* or facial or oral or craniofacial))).tw,kf.

37 (Lumpectom* or mastectom* or colostom* or Reconstructive surger* or Reconstruction).tw,kf.

38 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 or 25 or 26 or 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34 or 35 or 36 or 37

39 Body Image/ or Body dysmorphic disorders/

40 ((Body or self) adj2 (confidence or esteem or Image or concept or perception or dysmorphi* or dissatisfaction or shame or anxiety or concern or satisfaction)).tw,kf.

41 quality of life.ti,ab,kf. or self-concept/ or "quality of life"/

42 39 or 40 or 41

43 (Camouflage* or make-up or cosmetic* or beauty treatment*).tw,kf. or exp Cosmetics/

44 38 and 42 and 43

45 randomized controlled trial.pt.

46 controlled clinical trial.pt.

47 randomi?ed.ab.

48 placebo.ab.

49 drug therapy.fs.

50 randomly.ab.

51 trial.ab.

52 groups.ab.

53 or/45-52

54 exp animals/ not humans.sh.

55 53 not 54

56 44 and 55

PsycInfo (Ovid)

1. body image/ or body dysmorphic disorder/ or self esteem/ or self concept/ or self confidence/ or self criticism/ or self perception/ or "quality of life"/

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2. (((Body or self) adj2 (confidence or esteem or Image or concept or perception or dysmorphi* or dissatisfaction or shame or anxiety or concern or satisfaction)) or quality of life).ti,ab.
3. 1 or 2
4. exp skin disorders/
5. exp ECZEMA/
6. exp MELANOMA/
7. burns/
8. exp Eye Disorders/
9. exp STRABISMUS/
10. hypopituitarism/
11. exp AMPUTATION/
12. exp Prostheses/
13. plastic surgery/
14. exp MASTECTOMY/
15. exp COLOSTOMY/
16. 4 or 5 or 6 or 7 or 8 or 9 or 10 or 11 or 12 or 13 or 14 or 15
17. ((visible or face or facial* or maxillofacial or jaw) adj3 (injur* or damage* or scars or scarring or scarred or dermatos*)).tw,kf.
18. (keloid* or cicatrix or Acne or Eczema or psoriasis or Epidermolysis Bullosa or port wine stain* or h?emangioma$ or vitiligo or birth mark* or birthmark*).tw,kf.
19. (melanoma* or burns or burn or burned or alopecia or exophthalm* or graves disease or thyroid eye disease or strabismus or (misalign* adj3 eye*) or blepharoptosis or ectropion).tw,kf.
20. ((eye or eyelid or head or neck or mouth or oral) adj3 (neoplasm* or cancer* or tumo?r*)).tw,kf.
21. (Dwarfism or dwarf or small stature or gigantism or restricted growth or torticollis or dystonia or dupuytren* contracture*).tw,kf.
22. (Loss of limb* or amputee* or artificial limb*).tw,kf.
23. (appearance adj3 (abnormal or malformation or problem* or distress)).tw,kf.
24. (visibl* adj3 (differen* or disabilit*)).tw,kf.
25. (deformit* or deformed or disfigured or disfiguring or disfigurement).tw,kf.
26. (palatoschisis or hare lip* or harelip* or (cleft* adj (lip* or palate* or orofacial* or facial or oral or craniofacial))).tw,kf.
27. (Lumpectom* or mastectom* or colostom* or Reconstructive surger* or Reconstruction).tw,kf.
28. or/17-27
29. 16 or 28
30. cosmetic techniques/
31. (Camouflage* or make-up or cosmetic* or beauty treatment*).tw.
32. 30 or 31

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33. 3 and 29 and 32
34. double blind.tw.
35. random* assigned.tw.
36. control.tw.
37. 34 or 35 or 36
38. exp animals/ not humans.sh.
39. 37 not 38
40. 33 and 39

EMBASE (Ovid)

1. body image/ or body dysmorphic disorder/ or self esteem/ or self concept/ or "quality of life"/
2. (((Body or self) adj2 (confidence or esteem or Image or concept or perception or dysmorphi* or dissatisfaction or shame or anxiety or concern or satisfaction)) or quality of life).ti,ab,kw.
3. 1 or 2
4. exp *scar/
5. exp *face injury/
6. exp *craniofacial malformation/
7. exp *skin disease/
8. exp *psoriasis/
9. exp *eczema/
10. *acne vulgaris/
11. exp *congenital skin disease/
12. exp *vascular tumor/
13. exp *pigment disorder/
14. exp *melanoma/
15. *burn/ or *burn contracture/ or *electric burn/ or *eye burn/ or *face burn/
16. exp *alopecia/
17. exp *exophthalmos/
18. exp *strabismus/
19. exp *eyelid disease/
20. exp *mouth tumor/
21. exp *head cancer/
22. exp *"head and neck cancer"/ or exp *"head and neck tumor"/
23. *growth disorder/ or *dwarfism/ or *gigantism/ or *stunting/
24. exp *dystonia/
25. *Dupuytren contracture/
26. exp *amputation/

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27. *limb prosthesis/
28. *amputee/
29. exp *reconstructive surgery/
30. exp *mastectomy/
31. *colostomy/
32. exp *mouth disease/
33. ((visible or face or facial* or maxillofacial or jaw) adj3 (injur* or damage* or scars or scarring or scarred or dermatos*)).tw,kf.
34. (keloid* or cicatrix or Acne or Eczema or psoriasis or Epidermolysis Bullosa or port wine stain* or h?emangioma$ or vitiligo or birth mark* or birthmark*).tw,kf.
35. (melanoma* or burns or burn or burned or alopecia or exophthalm* or graves disease or thyroid eye disease or strabismus or (misalign* adj3 eye*) or blepharoptosis or ectropion).tw,kf.
36. ((eye or eyelid or head or neck or mouth or oral) adj3 (neoplasm* or cancer* or tumo?r*)).tw,kf.
37. (Dwarfism or dwarf or small stature or gigantism or restricted growth or torticollis or dystonia or dupuytren* contracture*).tw,kf.
38. (Loss of limb* or amputee* or artificial limb*).tw,kf.
39. (appearance adj3 (abnormal or malformation or problem* or distress)).tw,kf.
40. (visibl* adj3 (differen* or disabilit*)).tw,kf.
41. (deformit* or deformed or disfigured or disfiguring or disfigurement).tw,kf.
42. (palatoschisis or hare lip* or harelip* or (cleft* adj (lip* or palate* or orofacial* or facial or oral or craniofacial))).tw,kf.
43. (Lumpectom* or mastectom* or colostom* or Reconstructive surger* or Reconstruction).tw,kf.
44. or/33-43
45. or/4-32
46. 44 or 45
47. (Camouflage* or make-up or cosmetic* or beauty treatment*).tw,kw. or exp cosmetic/
48. 3 and 46 and 47
49. "randomized controlled trial (topic)"/
50. Randomized Controlled Trial/
51. Randomization/
52. Double Blind Procedure/
53. single blind procedure/
54. placebo/
55. (random* or sham or placebo*).ti,ab,hw.
56. ((singl* or doubl*) adj (blind* or dumm* or mask*)).ti,ab,hw.
57. ((tripl* or trebl*) adj (blind* or dumm* or mask*)).ti,ab,hw.
58. or/49-57

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59. 48 and 58

CINAHL

Search ID#	Search Terms
S66	S53 AND S65
S65	S54 OR S55 OR S56 OR S57 OR S58 OR S59 OR S60 OR S61 OR S62 OR S63 OR S64
S64	TX allocat* random*
S63	(MH "Quantitative Studies")
S62	(MH "Placebos")
S61	TX placebo*
S60	TX random* allocat*
S59	(MH "Random Assignment")
S58	TX randomi* control* trial*
S57	TX ((singl* n1 blind*) or (singl* n1 mask*)) or TX ((doubl* n1 blind*) or (doubl* n1 mask*)) or TX ((tripl* n1 blind*) or (tripl* n1 mask*)) or TX ((trebl* n1 blind*) or (trebl* n1 mask*)
S56	TX clinic* n1 trial*
S55	PT Clinical Trial
S54	(MH "Clinical Trials+")
S53	S43 AND S49 AND S52
S52	S50 OR S51
S51	TI ((Camouflage* or make-up or cosmetic* or beauty treatment*)) OR AB ((Camouflage* or make-up or cosmetic* or beauty treatment*))
S50	(MH "Cosmetics")
S49	S44 OR S45 OR S46 OR S47 OR S48
S48	(MH "Self Concept+")
S47	(MH "Quality of Life+") OR "quality of life"
S46	TI (((Body or self) N2 (confidence or esteem or Image or concept or perception or dysmorphi* or dissatisfaction or shame or anxiety or concern or satisfaction))) OR AB (((Body or self) N2 (confidence or esteem or Image or concept or perception or dysmorphi* or dissatisfaction or shame or anxiety or concern or satisfaction)))

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S45	(MH "Body Dysmorphic Disorder")
S44	(MH "Body Image+")
S43	S1 OR S2 OR S3 OR S4 OR S5 OR S6 OR S7 OR S8 OR S9 OR S10 OR S11 OR S12 OR S13 OR S14 OR S15 OR S16 OR S17 OR S18 OR S19 OR S20 OR S21 OR S22 OR S23 OR S24 OR S25 OR S26 OR S27 OR S28 OR S29 OR S30 OR S31 OR S32 OR S33 OR S34 OR S35 OR S36 OR S37 OR S38 OR S39 OR S40 OR S41 OR S42
S42	TI ((Lumpectom* or mastectom* or colostom* or Reconstructive surger* or Reconstruction)) OR AB ((Lumpectom* or mastectom* or colostom* or Reconstructive surger* or Reconstruction))
S41	TI ((palatoschisis or hare lip* or harelip* or (cleft* N1 (lip* or palate* or orofacial* or facial or oral or craniofacial)))) OR AB ((palatoschisis or hare lip* or harelip* or (cleft* N1 (lip* or palate* or orofacial* or facial or oral or craniofacial))))
S40	TI ((deformit* or deformed or disfigured or disfiguring or disfigurement)) OR AB ((deformit* or deformed or disfigured or disfiguring or disfigurement))
S39	TI ((visibl* N3 (differen* or disabilit*))) OR AB ((visibl* N3 (differen* or disabilit*)))
S38	TI ((appearance N3 (abnormal or malformation or problem* or distress))) OR AB ((appearance N3 (abnormal or malformation or problem* or distress)))
S37	TI ((Loss of limb* or amputee* or artificial limb*)) OR AB ((Loss of limb* or amputee* or artificial limb*))
S36	TI ((Dwarfism or dwarf or small stature or gigantism or restricted growth or torticollis or dystonia or dupuytren* contracture*)) OR AB ((Dwarfism or dwarf or small stature or gigantism or restricted growth or torticollis or dystonia or dupuytren* contracture*))
S35	TI (((eye or eyelid or head or neck or mouth or oral) N3 (neoplasm* or cancer* or tumo?r*))) OR AB (((eye or eyelid or head or neck or mouth or oral) N3 (neoplasm* or cancer* or tumo?r*)))
S34	TI ((melanoma* or burns or burn or burned or alopecia or exophthalm* or graves disease or thyroid eye disease or strabismus or (misalign* N3 eye*) or blepharoptosis or ectropion)) OR AB ((melanoma* or burns or burn or burned or alopecia or exophthalm* or graves disease or thyroid eye disease or strabismus or (misalign* N3 eye*) or blepharoptosis or ectropion))
S33	TI ((keloid* or cicatrix or Acne or Eczema or psoriasis or Epidermolysis Bullosa or port wine stain* or h?emangioma$ or vitiligo or birth mark* or birthmark*)) OR AB ((keloid* or cicatrix or Acne or Eczema or psoriasis or Epidermolysis Bullosa or port wine stain* or h?emangioma$ or vitiligo or birth mark* or birthmark*))
S32	TI (((visible or face or facial* or maxillofacial or jaw) N3 (injur* or damage* or scars or scarring or scarred or dermatos*))) OR AB (((visible or face or facial* or

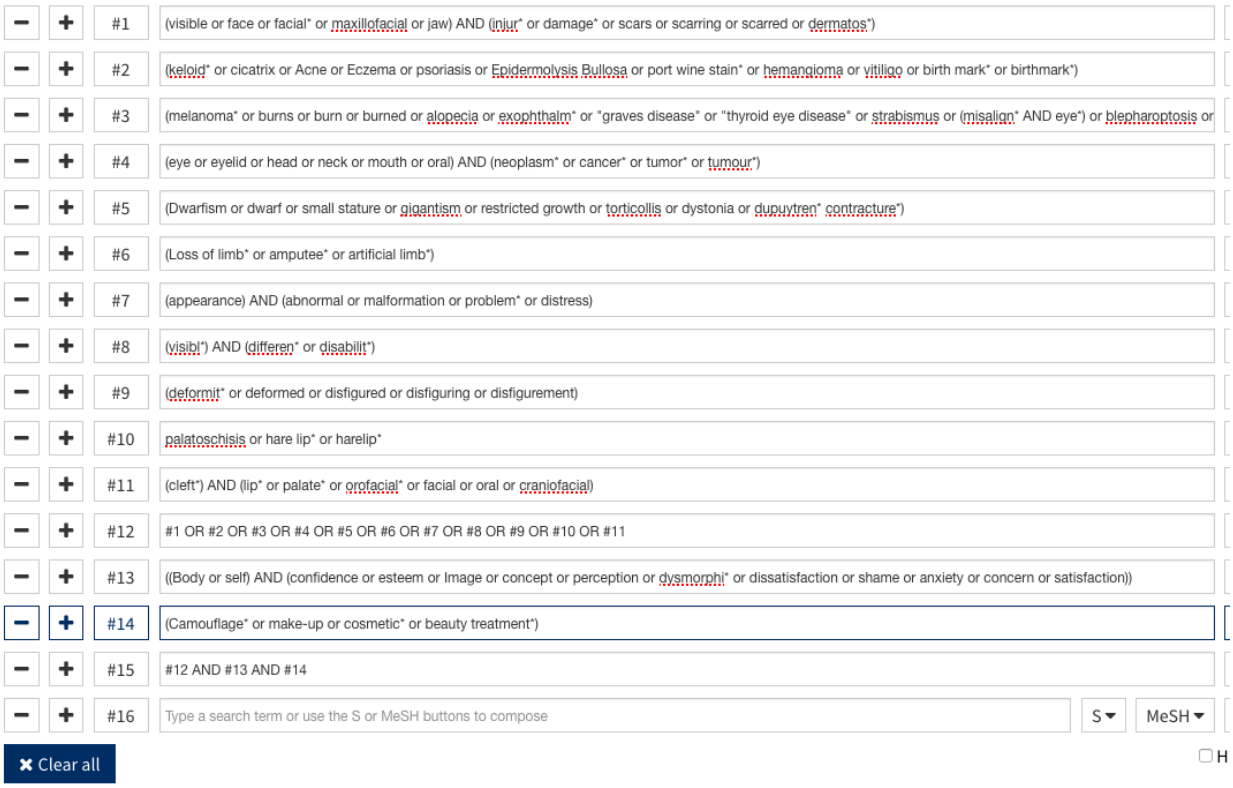
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	maxillofacial or jaw) N3 (injur* or damage* or scars or scarring or scarred or dermatos*)))
S31	(MH "Mouth Abnormalities+")
S30	(MH "Colostomy")
S29	(MH "Mastectomy+")
S28	(MH "Surgery, Reconstructive+")
S27	(MH "Amputees")
S26	(MH "Limb Deformities, Congenital+")
S25	(MH "Limb Prosthesis")
S24	(MH "Amputation+")
S23	(MH "Dupuytren's Contracture")
S22	(MH "Dystonia+")
S21	(MH "Beckwith-Wiedemann Syndrome") OR (MH "Lipodystrophy, Congenital Generalized") OR (MH "Sotos' Syndrome")
S20	(MH "Dwarfism+")
S19	(MH "Growth Disorders+")
S18	(MH "Head and Neck Neoplasms+")
S17	(MH "Mouth Neoplasms+")
S16	(MH "Eyelid Diseases+")
S15	(MH "Strabismus")
S14	(MH "Exophthalmos+")
S13	(MH "Alopecia+")
S12	(MH "Burns") OR (MH "Burns, Chemical") OR (MH "Burns, Electric")
S11	(MH "Melanoma+")
S10	(MH "Pigmentation Disorders+")
S9	(MH "Hemangioma+")
S8	(MH "Skin Abnormalities+")
S7	(MH "Acne Vulgaris")
S6	(MH "Eczema")
S5	(MH "Psoriasis")
S4	(MH "Facial Dermatoses+")

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S3	(MH "Craniofacial Abnormalities+")
S2	(MH "Facial Injuries+")
S1	(MH "Cicatrix+")

Cochrane Central Database of Controlled Trials



#1 (visible or face or facial* or maxillofacial or jaw) AND (injur* or damage* or scars or scarring or scarred or dermatos*)
#2 (keloid* or cicatrix or acne or eczema or psoriasis or Epidermolysis Bullosa or port wine stain* or hemangioma or vitiligo or birth mark* or birthmark*)
#3 (melanoma* or burns or burn or burned or alopecia or exophthalm* or "graves disease" or "thyroid eye disease" or strabismus or (misalign* AND eye*) or blepharoptosis or ectropion)
#4 (eye or eyelid or head or neck or mouth or oral) AND (neoplasm* or cancer* or tumor* or tumour*)
#5 (Dwarfism or dwarf or small stature or gigantism or restricted growth or torticollis or dystonia or dupuytren* contracture*)
#6 (Loss of limb* or amputee* or artificial limb*)
#7 (appearance) AND (abnormal or malformation or problem* or distress)
#8 (visibl*) AND (differen* or disabilit*)
#9 (deformit* or deformed or disfigured or disfiguring or disfigurement)
#10 (palatoschisis or hare lip* or harelip*
#11 (cleft*) AND (lip* or palate* or orofacial* or facial or oral or craniofacial)

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#12 #1 AND #2 AND #3 AND #4 AND #5 AND #6 AND #7 AND #8 AND #9 AND #10 AND #11
#13 (Body or self) AND (confidence or esteem or Image or concept or perception or dysmorphi* or dissatisfaction or shame or anxiety or concern or satisfaction)
#14 (Camouflage* or make-up or cosmetic* or beauty treatment*)
#15 #12 AND #13 AND #14

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APPENDIX 2: Variables Included in Data Extraction Form

Study characteristics

First author
Year
Journal
Funding source
Country
Medical condition
Recruitment setting and method
Number of participants randomised
Percent female
Mean (standard deviation [SD]) age
Mean (SD) years with body image concerns
Notes

Intervention and control group details

Group name
Description of intervention
Number of participants randomised to group
Number of participants analyzed in group
Notes

Outcomes

Outcome type (appearance-related, general psychological, reported harms)
Outcome measure name
Timepoints collected
Measure of aggregation (e.g., mean, median) at baseline and subsequent post-test assessments
Measure of dispersion (e.g., SD) at baseline and subsequent post-test assessments
Number of participants analyzed at baseline and subsequent post-test assessments
Other relevant results at baseline and subsequent post-test assessments
Notes

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	3-4
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	6-7
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	7
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	8
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	8
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	8-9
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Supp 1
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	9
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	9-10
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	10, Suppl 2
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	10
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	10-11
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	10-11

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	10
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	NA
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	12
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	28
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	33
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	29-32
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	14-15
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	15-16
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	NA
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	16
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	17-18
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	16-17, 18-19
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	4,26

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

For more information, visit: **www.prisma-statement.org**.

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BMJ Open

Effects of Cosmetic and Other Camouflage Interventions on Appearance-related and Psychological Outcomes among Adults with Visible Differences in Appearance: A Systematic Review

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Primary Subject Heading:	Complementary medicine
Secondary Subject Heading:	Evidence based practice
Keywords:	COMPLEMENTARY MEDICINE, EPIDEMIOLOGY, MENTAL HEALTH

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Effects of Cosmetic and Other Camouflage Interventions on Appearance-related and Psychological Outcomes among Adults with Visible Differences in Appearance: A Systematic Review

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ABSTRACT

Objective: Visible differences in appearance are associated with poor social and psychological outcomes. Effectiveness of non-surgical cosmetic and other camouflage interventions is poorly understood. The objective was to evaluate effects of cosmetic and other camouflage interventions on appearance-related outcomes, general psychological outcomes, and adverse effects for adults with visible appearance differences.

Design: Systematic review.

Data Sources: MEDLINE (Ovid), EMBASE (Ovid), PsycInfo (Ovid) CINAHL, and Cochrane Central databases searched from inception through October 24, 2020. Two reviewers independently reviewed titles and abstracts and full-texts.

Eligibility Criteria: Randomised controlled trials in any language on non-surgical cosmetic or other camouflage interventions that reported appearance-related outcomes, general psychological outcomes, or adverse effects for adults with visible appearance differences.

Data Extraction and Synthesis: Two reviewers independently extracted data, assessed intervention reporting using the Template for Intervention Description and Replication (TIDieR) checklist, and assessed risk of bias using the Cochrane risk of bias tool. Outcomes included appearance-related outcomes, general psychological outcomes (e.g., depression, anxiety), and adverse effects.

Results: One head-to-head trial and five trials with waiting list or routine care comparators were included. All had unclear or high risk of bias in at least five of seven domains. Effect sizes could not be determined for most outcomes due to poor reporting. Between-group statistically significant differences were not reported for any appearance-related outcomes and for only five

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of twenty-five (20%) other psychological outcomes. Given heterogeneity of populations and interventions, poor reporting, and high risk of bias, quantitative synthesis was not possible.

Conclusions: Conclusions about effectiveness of non-surgical cosmetic or other camouflage interventions could not be drawn. Well-designed and conducted trials are needed. Without such evidence, clinicians or other qualified individuals should engage with patients interested in cosmetic interventions in shared decision making, outlining potential benefits and harms, and the lack of evidence to inform decisions.

Funding Source: Canadian Institutes of Health Research (CMA – 151728).

Registration: PROSPERO (CRD42018103421).

Key Words: Appearance, body image, visible appearance differences, disfigurement, cosmetic, camouflage, systematic review.

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STRENGTHS AND LIMITATIONS OF THIS STUDY

- A peer-reviewed database search strategy was used to ensure rigour and comprehensiveness.
- The Template for Intervention Description and Replication (TIDieR) checklist was used to assess completeness of reporting of intervention characteristics.
- A patient advisory team was involved in reviewing and interpreting results and providing suggestions for implications.
- The outcome domains and measures reported by included trials were heterogeneous, which made comparison across trials difficult.
- The low quality of evidence available did not allow conclusions to be drawn about effectiveness.

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INTRODUCTION

Visible differences in physical appearance due to medical conditions or injuries are associated with poor social and mental health outcomes.[1-3] Individuals living with visible appearance differences experience unique social challenges, such as staring and invasive questions, and many experience fear of negative evaluation, social interaction anxiety, and social isolation.[1, 4] Intrapersonal consequences include body image dissatisfaction, low self-esteem, and high levels of emotional distress.[1-6]

Interventions that have been used to attempt to improve outcomes among people with visible appearance differences include surgical,[6, 7] psychological,[2, 8-10] and non-surgical cosmetic and other camouflage [11, 12] interventions. Non-surgical cosmetic and other camouflage interventions are comprised of non-invasive techniques intended to minimize or disguise visible appearance differences.[11, 12] These may include the use of cosmetics (e.g., concealer), prosthetics (e.g., prosthetic eye), clothing and stylistic accessories (e.g., scarves), or hair (e.g., wigs).[1]

A 2018 systematic review examined the effects of cosmetic camouflage in children and adults with disfiguring skin disorders on quality of life outcomes.[11] It included cosmetic interventions but not other types of camouflage interventions, and only included people with skin conditions but not appearance differences that were congenital, acquired from medical conditions, treatments, or injuries. Only 6 of 18 included studies were randomised controlled trials (RCTs), and the systematic review did not compare any outcomes between participants randomised to receive camouflage interventions and participants in control arms. Instead, the systematic reported only within-group pre-post mean reductions in Dermatological Life Quality Index scores for the nine studies with pre-post data that reported that measure.

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Pre-post analyses of intervention effects, however, are highly prone to bias, and when there is no control group it is not possible to determine with any degree of confidence whether an intervention had an effect or to attempt to quantify effects. This would require a precise understanding of the natural history of symptoms or to be able to confidently assume that symptoms do not change over time without intervention. This is typically not the case, however, in trials of interventions that attempt to improve mental health or other areas of subjective well-being. In these areas, participants often seek services and enrol in trials when they are experiencing high levels of symptoms or feeling poorly, and regression to the mean is common.[13-17] In major depression intervention trials, for instance, approximately 40% of participants assigned to placebo groups in drug trials or no-treatment groups in psychological intervention trials achieve remission, which has been termed "spontaneous recovery".[18] The Cochrane Handbook discourages inclusion of evidence from non-randomised studies when conducting trials is highly feasible and when evidence from non-randomised trials would be unlikely to provide interpretable results.[19]

We conducted a systematic review with the aim of informing health care providers and the public about the effectiveness of cosmetic and other camouflage interventions to improve outcomes among people with visible appearance differences or, in the absence of informative evidence, to identify this shortcoming and describe improvements that are needed. Thus, the objective was to evaluate evidence from RCTs of cosmetic and other camouflage interventions with training or instructions delivered (a) in person or (b) as self-help material (e.g., leaflets, internet-based) on appearance-related outcomes, general psychological outcomes, and adverse effects among adults with appearance differences that were congenital, from injury, a medical condition, or medical treatment.

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METHODS

This systematic review was registered in PROSPERO (CRD42018103421), and results were reported per the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement.[20]

Identification of Eligible Studies

Eligible studies were reports of RCTs in any language that examined the effects of non-surgical cosmetic or other camouflage interventions on appearance-related or general psychological outcomes among adults (ages 18 and over) with congenital or acquired visible appearance differences from injury (e.g., burn injury), disease or medical condition (e.g., scleroderma), or medical treatment (e.g., resection of facial tumor, breast reconstruction). Studies that included both children and adults were only included if results for the adults were reported separately or if greater than 80% of participants were adults. Trials that enrolled only individuals with visible appearance differences due to self-harm were excluded. Eligible interventions were defined as interventions with a core component that involved teaching, modelling, or demonstrating the modification or enhancement of some aspect of physical appearance through non-surgical cosmetic or other camouflage strategies. Eligible intervention formats included individual and group formats and delivery via any mode (e.g., in-person, web-based, bibliotherapy) by any type of provider. Eligible comparators included (1) any inactive control condition (e.g., usual care, attention control, sham procedure); (2) a non-cosmetic intervention designed to target appearance concerns (e.g., psychological intervention); and (3) another eligible cosmetic intervention. To be eligible for inclusion, studies must have reported (1) appearance-related psychological outcomes; (2) general psychological outcomes, including symptoms of depression or anxiety; or (3) adverse effects from the intervention.

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Search Strategy

A health sciences librarian (JB) developed the search strategy and performed the literature searches in MEDLINE (Ovid), EMBASE (Ovid), PsycInfo (Ovid) CINAHL, and Cochrane Central from database inception until October 24, 2020 with no language restrictions. The Cochrane Highly Sensitive Search Strategy for identifying randomised trials in MEDLINE: sensitivity-maximizing version (2008 revision) was used.[21] The MEDLINE strategy was developed with input from the project team and peer-reviewed by a second librarian using the Peer Review of Electronic Search Strategies (PRESS) guidelines.[22] After the initial MEDLINE strategy was finalized, it was adapted for use in the other databases. The complete search strategy is available in Supplementary File 1. We also reviewed reference lists of included articles and relevant reviews and conducted a forward citation search via Google Scholar.[23] We searched ClinicalTrials.gov (clinicaltrials.gov), ISRCTN (www.isrctn.com), and the World Health Organization registry search portal (apps.who.int/trialsearch) to identify any additional trials that may have been initiated but not published.

Search results were initially downloaded into the citation management database RefWorks (RefWorks, RefWorks-COS, Bethesda, MD, USA) for duplicate removal. Unique citation records were then transferred into the systematic review program DistillerSR (Evidence Partners, Ottawa, Canada). DistillerSR was used to track results of the review process. Two investigators independently reviewed titles and abstracts for eligibility. If either reviewer deemed a citation potentially eligible based on a review of the title and abstract, a full-text review of the article was completed by two reviewers independently. Any disagreement between reviewers after full-text review was resolved by consensus, including consultation with an independent third reviewer if necessary.

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Data Extraction and Analysis

Data were extracted from included RCTs by two investigators using a standardized electronic data collection form in DistillerSR. Disagreements were resolved by consensus and, if necessary, consultation with an independent third reviewer. See Supplementary File 2 for a complete list of extracted variables.

We used the Template for Intervention Description and Replication (TIDieR) checklist [24] to evaluate the degree to which interventions were reported sufficiently to allow replication in research or practice. The checklist is comprised of 12 items that assess reporting of the intervention name; the rationale or theory underlying the intervention; physical or informational material used; procedures and processes of the intervention; the intervention provider and their background; the mode of delivery, such as group or face-to-face; the location where the intervention was delivered and necessary infrastructure; the number of sessions, schedule, and duration; if any tailoring was done and how; any modifications made to the intervention; if adherence or fidelity was assessed and how; and, if adherence or fidelity was assessed, the extent to which the intervention was delivered as planned.

We used the 2011 version of the Cochrane Collaboration's risk of bias tool to evaluate risk of bias.[25] The tool has 7 domains, including random sequence generation, allocation concealment, blinding of participants and personnel, blinding of outcome assessment, incomplete outcome data, selective outcome reporting, and any other potential sources of bias. Studies were rated low, unclear, or high risk of bias on each domain.

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We did not pool results across trials because of the heterogeneity of interventions and trial designs and concerns about risk of bias. Rather, we descriptively reported trial characteristics and outcomes. Between groups standardized mean differences (SMDs) were calculated for each relevant outcome using Hedges's *g* [26] when possible. For trials where sufficient data were not provided to calculate SMDs, other relevant information that could be used to describe intervention outcomes (e.g., median change in each trial arm with p value) was reported.

Patient and Public Involvement

We assembled a Body Image Patient Advisory Team comprised of four female and one male persons with scleroderma, a rare disease that often causes changes in appearance, to advise on the present systematic review. The advisory team participated in a patient panel to discuss the study objectives, findings, and implications of the study. Members of the patient panel were also sent copies of the present manuscript prior to submission and asked to provide any feedback or suggestions. Members of the advisory team commented on issues related to characteristics of participants in the trials we reviewed, the acceptability of camouflage interventions, and delivery of these interventions via health care providers. These comments were incorporated into the Discussion section of the manuscript.

Modifications to Protocol

When we developed the review protocol, we did not anticipate that we might encounter trials that enrolled only individuals with visible appearance differences from self-inflicted injuries, and eligibility criteria did not address this group. We subsequently added visible appearance differences from self-injury as an exclusion criterion because we deemed the population to be substantively different than the target population; interventions for this

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population would need to address factors related to self-harming behaviour, in addition to appearance, which would be out of scope for our intended target population.

RESULTS

Search Results

The combined database searches identified 1,085 unique titles and abstracts. Of these, 1,062 were excluded after screening titles and abstracts and 19 after full-text review, resulting in 4 eligible RCTs. One additional eligible RCT [27] was identified via a database search that we conducted for a separate systematic review on psychological interventions (PROSPERO: CRD42018100632), and a second additional eligible RCT [28] was identified via manual searching of a relevant systematic review,[11] resulting in 6 included RCTs.[27-32] See Figure 1. No additional eligible trials were identified via manual searching of reference lists of included articles, citation tracking, or review of trial registries. One ongoing trial was identified via the review of trial registries (NCT03540966), but results were not available as of November 2, 2020.

Characteristics of Included Studies

The 6 included RCTs were published between 2006 and 2019.[27-32] Five trials compared an intervention to a no-intervention, waitlist, or usual care control,[27, 29-32] and one trial [28] was a head-to-head comparison of two different camouflage products.

Comparison to No Intervention or Usual Care

Trial participants included survivors of gynaecologic malignancies with chemotherapy-induced alopecia,[27] head and neck cancer survivors,[29] people living with acne vulgaris,[30] head and neck skin cancer survivors,[31] and survivors of breast cancer.[32] The number of participants per trial ranged from 24 to 136 (mean: 65;

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median: 50), and there was a total of 324 participants, with 162 participants each in intervention and control arms of trials. All participants were female. The mean ages for participants in each trial ranged from 29 to 67. See Table 1.

The trials were conducted in the United States,[27] Taiwan,[29] Japan,[30] Italy,[31] and Germany.[32] The interventions ranged from one to four sessions. Formats included videotape review,[27], combined individual and group sessions,[29] combined individual sessions and videotape,[30] and group sessions. [31, 32] Two RCTs required either self-report or investigator confirmation of appearance changes from an illness, injury, or medical procedure as a condition of eligibility;[29, 32] none, though, set a threshold for severity. No trials included patient report of concern about appearance as part of eligibility criteria, even though the trials targeted appearance-related and psychological variables.

In two RCTs, participants were provided with cosmetic camouflage products.[30, 31] In one trial,[30] which included people with acne, all products were developed and provided by Nov Pharmaceuticals and included skincare and cosmetic products and instructions on using the products that were specific to people with acne. In the second trial, customized camouflage plans that were first submitted to participants for approval were provided, and cosmetic camouflage products were provided by Istituto Ganassini Corporate;[31] however the specific cosmetic products were not described. Three of the trials [27, 29, 32] provided instructions on using camouflage techniques but did not provide specific products. In one trial,[27] participants viewed a 45-minute video that provided camouflage suggestions and techniques, including information on cosmetic camouflage and wigs. In another, hands-on instruction and practice with cosmetic camouflage was initiated, and a 25-page manual was provided.[29] In another, a group makeup workshop was provided and followed with a professional photoshoot and delivery of edited

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photographs.[32]

Comparators included standard counselling on hair loss,[27] unspecified routine care,[29, 31] conventional acne treatment,[30] and a wait-list control.[32] Four trials included appearance-related outcome measures,[27, 29, 31, 32] two trials included a depression symptom outcome measure,[29, 32] one trial included an anxiety outcome measure,[29] and five trials included general mental health outcome measures.[27, 29-32] No trials reported on adverse effects from participation.

Head-to-head Trial

The head-to-head trial,[28] which was conducted in France, included 88 healthy female volunteers with various facial aesthetic imperfections (e.g., melisma, scars from chicken pox), the presence of which were validated by clinical examination. The mean age of participants was 30. Participants were randomised to receive either a heavier-coverage liquid or a lighter-coverage powder product; all participants received recommendations and instructions from a professional makeup trainer in a single individual session. The study outcome was a body image-specific quality of life measure. See Table 1.

Synthesis of Outcomes

Comparison to No Intervention or Routine Care Controlled Trials

There was sufficient information to calculate intervention effect sizes (SMDs) and confidence intervals for two of the five trials that compared an intervention group to a no intervention, wait list, or routine care control group,[29, 30] only one of which [29] included an appearance-related outcome.

Appearance-related Outcomes: In the four trials that reported appearance-related

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outcomes,[27, 29, 31, 32] there were no statistically significant differences between camouflage intervention and control groups. Reported effect sizes were generally negligible. See Table 2.

General Psychological Symptoms: Across a total of 25 general psychological outcome measurement points reported in the five trials, five (20%) were statistically significant. There was a small to moderate effect on the Leibowitz Social Anxiety Scale-Anxiety subscale at 8 weeks (SMD: -0.41 CI: -0.89, -0.07) in a trial that randomised 70 head and neck cancer survivors to individual and group sessions or routine care,[29] although the effect was smaller and not statistically significant at 4 or 12 weeks, and the effect for the Fear subscale was non-significant at all three time points. There were moderate to large magnitude effects for depressive symptoms (*d*: -0.83 at 2 weeks and *d*: -0.43 at 4 weeks) and self-esteem (*d*: 0.68 at 2 weeks and *d*: 0.72 at 4 weeks) in a trial that randomised 44 breast cancer patients to a single group session versus wait-list control, though it was not clear if they were statistically significant.[32] See Table 2.

Head-to-head Trial

In the head-to-head trial, which compared liquid and powder camouflage products among 88 people with a variety of aesthetic imperfections,[28] sufficient information to calculate effect sizes was not provided. Of the four outcomes measured at two time points (8 outcomes total), one was reported to be statistically significantly different between groups (a self-confidence subscale at 1 week).

Reporting of Intervention Components and Risk of Bias

As assessed using the TIDieR checklist, few trial reports described the interventions used with sufficient detail to fully understand what had occurred or to consider replication (See Table 3). Materials were not sufficiently reported in terms of describing specific materials used or providing details on how to access the materials for all of the trials.[27-32] Although all of the

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studies described who provided the intervention, four did not provide details about training or qualifications of the interventionists specific to camouflage or visible appearance differences.[28, 30-32] Adherence was not assessed in any of the six trials.

Risk of bias was evaluated using the Cochrane criteria with most studies rated as high or unclear across the domains assessed (See Table 4). Random sequence generation procedures were only described in two of the studies.[29, 31] Due to the nature of the interventions and self-reported psychological outcomes used in the studies, risk of bias due to a lack of blinding of participants/personnel and outcome assessment was high across all studies. Overall, of the seven domains assessed (including "other"), three RCTs were rated as unclear or high in five domains,[27, 29, 32] two in six domains,[30, 31] and one in all seven domains.[28]

DISCUSSION

Main Findings

We identified five RCTs that compared camouflage interventions to no treatment, waiting list, or routine care controls, and one head-to head trial that compared two camouflage products. Trials included participants from populations with a range of visible appearance differences, including people with head and neck cancer, acne vulgaris, facial skin cancer, chemotherapy-induced alopecia, breast cancer, and non-defined facial aesthetic imperfections. Risk of bias was unclear or high in at least five of seven possible domains for all included RCTs. Furthermore, outcomes were incompletely reported in many trials, and we could only calculate effect sizes for a small proportion of outcomes that were described. Very few statistically significant results were identified. Additionally, based on TIDieR checklist ratings, interventions were generally not described with enough information to support replication or use in practice. Given these limitations, a meaningful quantitative synthesis of the findings from included RCTs was not

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possible, and conclusions about the likely effectiveness of non-surgical cosmetic interventions on appearance-related or general psychological outcomes could not be drawn with any degree of confidence. No trials reported on adverse effects.

Findings in Context

A previous review [11] used pre- and post-intervention scores among only studies that used a single quality of life measure, the Dermatology Life Quality Index (DLQI),[33] and concluded that reductions in scores across nine studies of varying designs suggested effectiveness of cosmetic camouflage on quality of life outcomes. That review did not attempt to synthesize between-group effects or describe any limitations due to not evaluating between-group effects. There are several reasons that help explain why our conclusion differs from the conclusion of that review. First, the specific focus of the previous review was the role of cosmetic camouflage on quality of life outcomes, whereas the present review was broader in scope and included any non-surgical cosmetic intervention (e.g., wigs, prosthetics) and any appearance-related and general psychological outcomes. Second, the previous review included studies of children and adults, whereas our review was limited to adult samples. Third, and most importantly, the conclusions of the previous review were drawn from pre-post comparisons without control groups, whereas we included only RCTs. RCTs are considered the gold standard study design for establishing the effectiveness of psychological interventions and inferring causation.[34] There is a high risk in trials of psychosocial interventions, in which people often consent to participate when they are struggling with concerns, that the natural trajectory of included participants may be to improve over time. Non-randomised trials in this area are pre-post trials that must assume that there would be no change over time without the intervention. Systematic reviews that draw conclusions and make recommendations from these kinds of trials

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are at high risk of misleading clinicians who seek to provide the best possible care to people with visible differences in appearance, misleading people with visible appearance differences about their options, and losing the opportunity to encourage researchers to engage in improved trials that answer a critical health care question.

Limitations

Several important methodological limitations were noted in the RCTs included in the present review, and there are implications for future research. First, study aims were often ambiguous, and it was unclear if what was being tested was a specific product, camouflage in general, or instructions on using camouflage. Second, few trials provided information that would be needed for replication or use in practice. Third, no studies had eligibility criteria for baseline appearance concerns or subjective severity. Only three of the trials [28, 29, 32] included eligibility criteria relating to confirming the presence of appearance differences; however, no trials included a threshold for objective severity or required that participants had appearance-related concerns. Thus, some participants may have been enrolled even though they did not have appearance concerns, and those participants would not have been able to benefit from the interventions. Fourth, none of the studies reported on harms from the interventions. It is possible that some products may exacerbate skin conditions and lead to worse objective appearance or psychological outcomes. Fifth, no studies included men, limiting generalizability of the findings. This was a limitation that was emphasized by the Patient Advisory Team, which raised the concern that the acceptability of camouflage among men is not known. Sixth, the small number of RCTs available, the generally small sample sizes included, and the poor reporting of trial methods and results limited utility of the trials and the ability to draw conclusions from what

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they reported; these are common issues in rare disease and visible appearance differences research.[35]

The findings of this review should be interpreted in the context of several limitations of the review itself. First, although a systematic search designed by a medical librarian was utilized, incompleteness of the evidence reported and publication bias limiting null results are concerns for any systematic review. Second, the small number of studies eligible for inclusion in the review and differences in how data were reported did not allow for publication bias to be assessed. Third, given the low quality of trial evidence available, in addition to not being able to draw conclusions about overall effectiveness, examination of potentially relevant participant (e.g., age, sex, marital status) and condition (e.g., years of living with the visible appearance difference, patient-perceived and observer-rated severity) variables could not be undertaken. Fourth, even with better quality evidence, the inconsistency across studies in outcome domains and measures used to evaluate effects would make drawing conclusions difficult. Ideally, a core set of outcome domains and measures would be used consistently.

Implications of Findings

There are clinical implications of this review. Because of high patient interest in non-surgical cosmetic treatments [12] and the fact that for some conditions causing visible appearance differences non-surgical cosmetic treatments are the only treatment option available, many people may consider using camouflage or other cosmetic interventions. A concern raised by members of our Body Image Patient Advisory Team, however, was that it may be better to accept one's visible appearance differences rather than engaging in behaviours to attempt to hide or conceal them. It was noted that suggesting camouflage may send an undesired message to patients who are comfortable with their appearance or who are working to be more comfortable.

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In addition to improving the quality and reporting of trials, future researchers may consider the role of internet delivery of interventions, as health interventions are increasingly delivered via the internet and have been shown to be similarly effective to face-to-face interventions.[36-38] Online delivery could enable wider recruiting and larger sample sizes as well as more efficient dissemination if an intervention is found to be effective.

Patients who are interested in these approaches and may wish to discuss advantages and disadvantages with health care providers or other qualified individuals. Given the lack of high-quality evidence, clinicians are left with a gap not only around whether interventions are likely to be effective, but whether there may be adverse effects to consider. In light of these limitations, we recommend that clinicians working with visible appearance differences populations engage in a shared decision making approach with patients who express an interest in or who may benefit from non-surgical cosmetic intervention,[39] such that education is provided on the current lack of empirical information about clinical and psychological harms as well as effectiveness of interventions.

Conclusion

In summary, the present review did not find evidence to evaluate benefits or harms of the use of non-surgical cosmetic treatments in visible appearance differences. There are several recommendations for future RCTs evaluating the efficacy of non-surgical cosmetic interventions in visible appearance differences populations, such as the inclusion of appearance-related eligibility criteria and the use of internet-based and other feasible delivery modalities. Until more robust scientific evidence becomes available, clinicians can engage in shared decision making with patients to determine whether engaging in non-surgical cosmetic interventions given the current lack of empirical support may be warranted.

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Figure legend:

Figure 1. PRISMA flow diagram of study selection

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AUTHORS' CONTRIBUTIONS

SG, DBR, ACJ, and BDT were responsible for the study concept and design, drafted the study protocol, contributed to data extraction, contributed to drafting the manuscript, and approved the final manuscript. JB designed the search strategies, AK conducted the search strategies, and both contributed to drafting the manuscript and approved the final manuscript. VLM and LK were responsible for the study concept and design, drafted the study protocol, contributed to drafting the manuscript, and approved the final manuscript. BDT is the guarantor.

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COMPETING INTERESTS STATEMENT

All authors have completed the ICMJE uniform disclosure form at www.icmje.org/coi_disclosure.pdf and declare that they have no competing interests.

DATA SHARING STATEMENT

All data extracted and analysed for the present study are reported in the manuscript and its tables.

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Table 1. Characteristics of included studies

Author Year	N Randomised	Country	Population	Age mean (sd)	Format	Materials Provided to Participants	Number of Sessions	Length of Session(s)	Comparator
No Intervention or Routine Care Controlled Trials									
Chen 2016 [29]	Total: 70 Tx: 35 Control: 35	Taiwan	Head and neck cancer survivors	52 (2)	Individual and group: Led by research nurses, a research cosmetologist, and cosmetology assistants	Educational manual	4 sessions (3 individual; 1 group)	Up to 3 hours	Routine care
Matsuoka 2006 [30]	Total: 50[a] Tx: 25 Control: 25	Japan	People with acne vulgaris	25 (4)	Individual and videotape: Led by patient's primary dermatologist with taped instructions	Unspecified camouflage products	1 session	Not reported	Standard acne treatment
Nicoletti 2014 [31]	Total: 24 Tx: 12 Control: 12	Italy	People with facial skin cancer	67 (NR)	Group: Led by professional makeup artist; psychologist also in attendance	Unspecified camouflage products	3 sessions	2 hours	Routine care
Nolte 2006 [27]	Total: 136 Tx: 68 Control: 68	United States	People with chemotherapy-induced alopecia	58 (12)[b]	Videotape: cosmetic camouflage and wig/hair covering techniques	No products provided	1 session	45 minutes	Standard counselling on hair loss
Richard 2018 [32]	Total: 44 Tx: 22 Control: 22	Germany	People with breast cancer	39 (9)	Group: makeup workshop, photoshoot, and receipt of edited photos	Unspecified camouflage products used in session but not provided; edited photographs	1 session	4 hours	Wait-list control
Head-to-Head Trial									
Beresniak 2017 [28]	Total: 88[c]	France	People with facial aesthetic imperfections	30 (8)	Individual: Recommendations and instructions provided by professional makeup trainer	Liquid or powder camouflage products	1 session	Not reported	Other camouflage product

[a]Number randomised not provided, authors noted that 50 patients participated in the study; [b]This is the age for N = 187 women in broader sample from which N=136 were drawn; [c]Number per group not provided; authors noted that 88 patients were randomised.

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Table 2. Outcomes of Included Trials. Scores reversed for some outcome measures such that higher scores indicate worse functioning for all measures and negative between-group effects reflect better outcomes in the intervention groups.

Study	N randomised	Appearance-related	General Mental Health	Depression	Social Anxiety
No Intervention or Routine Care Controlled Trials					
Chen 2016 [29]	Total: 70[a] Tx: 35 Control: 35	BIS: SMD (95% CI) 4 weeks -0.07 (-0.54, 0.40) 8 weeks -0.08 (-0.56, 0.39) 12 weeks -0.43 (-0.92, 0.06)	RSES: SMD (95% CI) 4 weeks -0.19 (-0.67, 0.28) 8 weeks 0.31 (-0.17, 0.79) 12 weeks -0.03 (-0.52, 0.45)	HADS-D: SMD (95% CI) 4 weeks 0.33 (-0.15, 0.80) 8 weeks -0.08 (-0.56, 0.39) 12 weeks -0.14 (-0.62, 0.35)	LSAS-F: SMD (95% CI) 4 weeks -0.29 (-0.76, 0.19) 8 weeks -0.29 (-0.77, 0.19) 12 weeks -0.41 (-0.89, 0.08) LSAS-A: SMD (95% CI) 4 weeks -0.32 (-0.80, 0.15) 8 weeks -0.41 (-0.89, -0.07) 12 weeks -0.33 (-0.82, 0.16)
Matsuoka 2006 [30]	Total: 50[b] Treatment: 25 Control: 25	NA	WHOQOL-26-P: SMD (95% CI) 4 weeks 0.05 (-0.50, 0.60) WHOQOL-26-SR: SMD (95% CI) 4 weeks -0.02 (-0.57, 0.54) DLQI-SF: SMD (95% CI) 4 weeks 0.00 (-0.55, 0.55) DLQI-PR: SMD (95% CI) 4 weeks	NA	NA

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Study	N randomised	Appearance-related	General Mental Health	Depression	Social Anxiety
			0.16 (-0.40, 0.71)		
Nicoletti 2014[c] [31]	Total: 24[d] Treatment: 12 Control: 12	BUT-GSI: 3 months[e]: *Tx:* change = -0.26, IQR = -0.72, 0.16 *Control:* change = 0.03, IQR = -0.08, 0.34 p = 0.371 6 months[e]: *Tx:* change = -0.15, IQR = -0.54, 0.45 *Control:* change = 0.12, IQR = -0.09, 0.64 p = 0.318	SP-PF: 3 months[e]: *Tx:* change = -4.75, IQR = -12.47, 5.02, *Control:* change = 7.15, IQR = 4.4-8.1 p = 0.270 6 months[e]: *Tx* change = -9.15, IQR = -10.3, -3.25 *Control* change = 18.1, IQR = -4.5, 28.33 p = 0.003 SP-SF: 3 months[e]: *Tx:* change = -3.54, IQR = -11.04, 17.02 *Control:* change = 13.84, IQR = -3.75, 17.92, p = 0.318 6 months[e]: *Tx group* change = -1.99, IQR = -16.00, 9.54 *Control group* change = 0.83, IQR = -5.17, 5.50 p = 0.636	NA	NA
Nolte 2006[h] [27]	Total: 136 Treatment: 68 Control: 68[i]	BCSCS-BC: Post-tx: F[1,132] = 0.11, p = 0.74 (Tx: 68; Control 68)	BCSCS-SC: Post-tx F[1,132] = 0.32, p = 0.57 (Tx: 68; Control 68)	NA	NA

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Study	N randomised	Appearance-related	General Mental Health	Depression	Social Anxiety
Richard 2018 [32]	Total: 44 Treatment: 22 Control: 22[f]	BIS: 2 weeks, 4 weeks: NR; Improvements in body image were independent of treatment (p > 0.05)	RSES: 2 weeks: *d* = -0.68[g] 4 weeks: *d* = -0.72[g]	CES-D: 2 weeks: *d* = -0.83[g] 4 weeks: *d* = -0.45[g]	NA
Head-to-head trial					
Beresniak 2017 [28]	Total: 88[j]	BeautyQOL-Attractiveness: at 1 and 3 weeks, mean A > mean B; p>0.05 BeautyQOL-Social Life: at 1 and 3 weeks, mean A > mean B; p>0.05 BeautyQOL-Mood: at 1 and 3 weeks, mean A > mean B; p >0.05 BeautyQOL-Self Confidence: 1 week: mean A > mean B; p <0.05 3 weeks: mean A > mean B; p >0.05	NA	NA	NA

[a]Tx (*n*: 35, 34, 32) and control (*n*: 34, 34, 34) at 4, 8, and 12 weeks, respectively; [b]Number randomised not provided, noted that 50 patients participated in the study; [c]Effect sizes could not be calculated; [d]Tx (*n*: 12, 12) and control (*n*: 12, 12) at 3 and 6 months, respectively; [e]Data reported as change score (IQR) from baseline to 3 months and p-value is for between-group difference; [f]Tx (*n*: 20, 29) and control (*n*: 19, 19) at 2 and 4 weeks; [g]Baseline corrected; [h]Intervention effect comparing baseline scores to scores after course 4 of chemotherapy using BCSCS scores during course 3 of chemotherapy as a covariate reported; [i]Tx (*n*: 68) and control (*n*: 68) at post-tx measurement; [j]Tx and Control *n* not provided; 86 participants completed study

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Abbreviations: BCSCS-BC (Body Cathexis/Self-Cathexis Scale: Body Cathexis); BCSCS-SC: Body Cathexis/Self-Cathexis Scale: Self-Cathexis; BIS: Body Image Scale (higher scores indicate poorer body image); BIS: Body Image Scale (higher scores indicate poorer body image); BUT-GSI: Body Uneasiness Test Global Score Index (higher scores indicate greater body uneasiness and negative change score indicates improvement); DLQI-PR (Dermatology Life Quality Index: Personal Relationships); DLQI-SF (Dermatology Life Quality Index: Symptoms/Feelings); HADS-D: Hospital Anxiety and Depression Scale: Depression subscale (higher scores indicate greater depression); LSAS-A: Liebowitz Social Anxiety Scale: Anxiety subscale (higher scores indicate greater anxiety); LSAS-F: Liebowitz Social Anxiety Scale: Fear subscale (higher scores indicate greater fear); RSES: Rosenberg Self-Esteem Scale); SP-PF: Satisfaction Profile: Psychological Functionality; SP-SF: Satisfaction Profile: Social Functionality; Tx = Treatment/intervention group; WHOQOL-26-P: World Health Organization Quality of Life: Psychological domain; WHOQOL-26-SR: World Health Organization Quality of Life: Social Relationships domain

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Table 3. TIDieR Checklist Results

Studies	Name	Why	Materials	Procedures	Who provided	How	Where	When and how much	Tailoring	Modifications	How well: Planned adherence	How well: Actual adherence
No Intervention or Routine Care Controlled Trials												
Chen 2016[29]	Yes	Yes	Partial[d]	Yes	Yes	Yes	No	Partial[e]	N/A	N/A	N/A	N/A
Matsuoka 2006[30]	Yes	Yes	Partial[d]	Yes	Partial[b]	Yes	Partial[f]	No	Yes	N/A	N/A	N/A
Nicoletti 2014[31]	Yes	Yes	Partial[g]	No	Partial[b]	Yes	Yes	Yes	Partial[c]	N/A	N/A	N/A
Nolte 2006[27]	Yes	Partial[h]	Partial[d]	Yes	Yes	Partial[i]	Partial[f]	Partial[j]	N/A	N/A	N/A	N/A
Richard 2018[32]	Yes	Yes	Partial[a]	Yes	Partial[b]	Partial[k]	Partial[f]	Yes	N/A	N/A	N/A	N/A
Head-to-head trial												
Beresniak 2017 [28]	Yes	Yes	Partial[a]	Yes	Partial[b]	No	No	Yes	Partial[c]	N/A	N/A	N/A
Total	6 Yes 0 Partial 0 No 0 N/A	5 Yes 1 Partial 0 No 0 N/A	0 Yes 6 Partial 0 No 0 N/A	5 Yes 0 Partial 1 No 0 N/A	2 Yes 4 Partial 0 No 0 N/A	3 Yes 2 Partial 1 No 0 N/A	1 Yes 3 Partial 2 No 0 N/A	3 Yes 2 Partial 1 No 0 N/A	1 Yes 2 Partial 0 No 3 N/A	0 Yes 0 Partial 0 No 6 N/A	0 Yes 0 Partial 0 No 6 N/A	0 Yes 0 Partial 0 No 6 N/A

Note: Yes (Y), No (N), Partial (P), Not applicable (N/A); [a]Specific brand information not provided; [b]Training information not provided; [c]Details regarding how adaptations were made not provided; [d]Information on how to access materials not provided; [e]Duration of all sessions not provided; [f]Details about infrastructure and other relevant feature not provided; [g]Specific products not described; [h]details about rationale underlying intervention not provided; [i]unclear if entire sessions was individual format; [j]Timeline of intervention provision unclear; [k]Number of participants per group/number of groups not specified .

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Table 4. Quality assessment for included RCTs

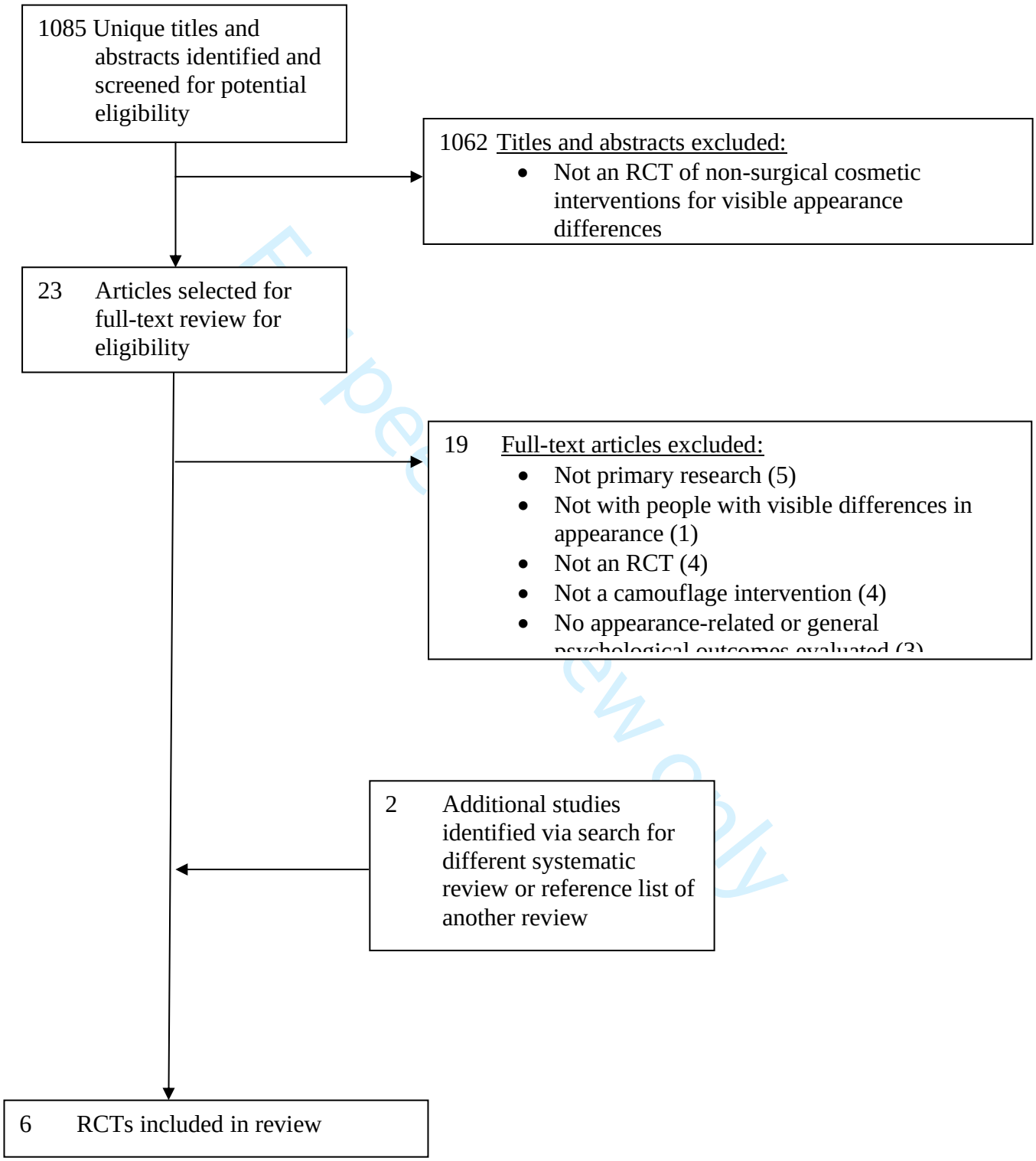
Studies	Random sequence generation	Allocation concealment	Blinding of participants/ personnel	Blinding outcome assessment	Incomplete outcome data	Selective reporting	Free of other bias
No Intervention or Routine Care Controlled Trials							
Chen 2016[29]	Low	Unclear	High	High	Low	Unclear	Unclear[b]
Matsuoka 2006[30]	Unclear	Unclear	High	High	Low	Unclear	Unclear[c]
Nicoletti 2014[31]	Low	Unclear	High	High	Unclear	Unclear	Unclear[c]
Nolte 2006[27]	Unclear	Unclear	High	High	Low	Unclear	Low
Richard 2018[32]	Unclear	Unclear	High	High	Low	Unclear	Low
Head-to-head trial							
Beresniak 2017 [28]	Unclear	Unclear	High	High	High	Unclear	Unclear[a]
Total	2 Low 4 Unclear 0 High	0 Low 6 Unclear 0 High	0 Low 0 Unclear 6 High	0 Low 0 Unclear 6 High	4 Low 1 Unclear 1 High	0 Low 6 Unclear 0 High	2 Low 4 Unclear 0 High

Note. Green indicates low risk of bias; yellow indicates unclear risk of bias; red indicates high risk of bias [a]Only mean scores reported, unclear if product brand and consultant affiliation are related; [b]Significant difference at baseline for experimental group in radiotherapy dosage[c] not controlled for in analysis; [c]All products provided by a single pharmaceutical company.

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Figure 1. PRISMA flow diagram of study selection



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Appendix 1. Search Strategies

Ovid MEDLINE(R) and Epub Ahead of Print, In-Process & Other Non-Indexed Citations, Ovid MEDLINE(R) Daily 1946 to Present

1 exp Cicatrix/
2 exp Facial Injuries/ or exp Craniofacial Abnormalities/ or exp Facial Dermatoses/
3 Psoriasis/ or Eczema/ or Acne Vulgaris/
4 exp Skin Abnormalities/
5 exp Hemangioma/
6 exp Pigmentation Disorders/
7 exp Melanoma/
8 Burns, Chemical/ or Burns/ or Burns, Inhalation/ or Burns, Electric/
9 exp Alopecia/
10 exp Exophthalmos/
11 exp Strabismus/
12 exp Eyelid Diseases/
13 exp Mouth Neoplasms/
14 exp "head and neck neoplasms"/
15 Growth Disorders/
16 exp Dwarfism/
17 Gigantism/
18 exp Dystonia/
19 Dupuytren Contracture/
20 Amputation/
21 Artificial Limbs/
22 Amputees/
23 exp reconstructive surgical procedures/
24 exp Mastectomy/
25 Colostomy/
26 exp Mouth Abnormalities/
27 ((visible or face or facial* or maxillofacial or jaw) adj3 (injur* or damage* or scars or scarring or scarred or dermatos*)).tw,kf.
28 (keloid* or cicatrix or Acne or Eczema or psoriasis or Epidermolysis Bullosa or port wine stain* or h?emangioma$ or vitiligo or birth mark* or birthmark*).tw,kf.
29 (melanoma* or burns or burn or burned or alopecia or exophthalm* or graves disease or thyroid eye disease or strabismus or (misalign* adj3 eye*) or blepharoptosis or ectropion).tw,kf.

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30 ((eye or eyelid or head or neck or mouth or oral) adj3 (neoplasm* or cancer* or tumo?r*)).tw,kf.
31 (Dwarfism or dwarf or small stature or gigantism or restricted growth or torticollis or dystonia or dupuytren* contracture*).tw,kf.
32 (Loss of limb* or amputee* or artificial limb*).tw,kf.
33 (appearance adj3 (abnormal or malformation or problem* or distress)).tw,kf.
34 (visibl* adj3 (differen* or disabilit*)).tw,kf.
35 (deformit* or deformed or disfigured or disfiguring or disfigurement).tw,kf.
36 (palatoschisis or hare lip* or harelip* or (cleft* adj (lip* or palate* or orofacial* or facial or oral or craniofacial))).tw,kf.
37 (Lumpectom* or mastectom* or colostom* or Reconstructive surger* or Reconstruction).tw,kf.
38 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 or 25 or 26 or 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34 or 35 or 36 or 37
39 Body Image/ or Body dysmorphic disorders/
40 ((Body or self) adj2 (confidence or esteem or Image or concept or perception or dysmorphi* or dissatisfaction or shame or anxiety or concern or satisfaction)).tw,kf.
41 quality of life.ti,ab,kf. or self-concept/ or "quality of life"/
42 39 or 40 or 41
43 (Camouflage* or make-up or cosmetic* or beauty treatment*).tw,kf. or exp Cosmetics/
44 38 and 42 and 43
45 randomized controlled trial.pt.
46 controlled clinical trial.pt.
47 randomi?ed.ab.
48 placebo.ab.
49 drug therapy.fs.
50 randomly.ab.
51 trial.ab.
52 groups.ab.
53 or/45-52
54 exp animals/ not humans.sh.
55 53 not 54
56 44 and 55

PsycInfo (Ovid)

1. body image/ or body dysmorphic disorder/ or self esteem/ or self concept/ or self confidence/ or self criticism/ or self perception/ or "quality of life"/

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2. (((Body or self) adj2 (confidence or esteem or Image or concept or perception or dysmorphi* or dissatisfaction or shame or anxiety or concern or satisfaction)) or quality of life).ti,ab.

3. 1 or 2

4. exp skin disorders/

5. exp ECZEMA/

6. exp MELANOMA/

7. burns/

8. exp Eye Disorders/

9. exp STRABISMUS/

10. hypopituitarism/

11. exp AMPUTATION/

12. exp Prostheses/

13. plastic surgery/

14. exp MASTECTOMY/

15. exp COLOSTOMY/

16. 4 or 5 or 6 or 7 or 8 or 9 or 10 or 11 or 12 or 13 or 14 or 15

17. ((visible or face or facial* or maxillofacial or jaw) adj3 (injur* or damage* or scars or scarring or scarred or dermatos*)).tw,kf.

18. (keloid* or cicatrix or Acne or Eczema or psoriasis or Epidermolysis Bullosa or port wine stain* or h?emangioma$ or vitiligo or birth mark* or birthmark*).tw,kf.

19. (melanoma* or burns or burn or burned or alopecia or exophthalm* or graves disease or thyroid eye disease or strabismus or (misalign* adj3 eye*) or blepharoptosis or ectropion).tw,kf.

20. ((eye or eyelid or head or neck or mouth or oral) adj3 (neoplasm* or cancer* or tumo?r*)).tw,kf.

21. (Dwarfism or dwarf or small stature or gigantism or restricted growth or torticollis or dystonia or dupuytren* contracture*).tw,kf.

22. (Loss of limb* or amputee* or artificial limb*).tw,kf.

23. (appearance adj3 (abnormal or malformation or problem* or distress)).tw,kf.

24. (visibl* adj3 (differen* or disabilit*)).tw,kf.

25. (deformit* or deformed or disfigured or disfiguring or disfigurement).tw,kf.

26. (palatoschisis or hare lip* or harelip* or (cleft* adj (lip* or palate* or orofacial* or facial or oral or craniofacial))).tw,kf.

27. (Lumpectom* or mastectom* or colostom* or Reconstructive surger* or Reconstruction).tw,kf.

28. or/17-27

29. 16 or 28

30. cosmetic techniques/

31. (Camouflage* or make-up or cosmetic* or beauty treatment*).tw.

32. 30 or 31

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33. 3 and 29 and 32
34. double blind.tw.
35. random* assigned.tw.
36. control.tw.
37. 34 or 35 or 36
38. exp animals/ not humans.sh.
39. 37 not 38
40. 33 and 39

EMBASE (Ovid)

1. body image/ or body dysmorphic disorder/ or self esteem/ or self concept/ or "quality of life"/
2. (((Body or self) adj2 (confidence or esteem or Image or concept or perception or dysmorphi* or dissatisfaction or shame or anxiety or concern or satisfaction)) or quality of life).ti,ab,kw.
3. 1 or 2
4. exp *scar/
5. exp *face injury/
6. exp *craniofacial malformation/
7. exp *skin disease/
8. exp *psoriasis/
9. exp *eczema/
10. *acne vulgaris/
11. exp *congenital skin disease/
12. exp *vascular tumor/
13. exp *pigment disorder/
14. exp *melanoma/
15. *burn/ or *burn contracture/ or *electric burn/ or *eye burn/ or *face burn/
16. exp *alopecia/
17. exp *exophthalmos/
18. exp *strabismus/
19. exp *eyelid disease/
20. exp *mouth tumor/
21. exp *head cancer/
22. exp *"head and neck cancer"/ or exp *"head and neck tumor"/
23. *growth disorder/ or *dwarfism/ or *gigantism/ or *stunting/
24. exp *dystonia/
25. *Dupuytren contracture/
26. exp *amputation/

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27. *limb prosthesis/
28. *amputee/
29. exp *reconstructive surgery/
30. exp *mastectomy/
31. *colostomy/
32. exp *mouth disease/
33. ((visible or face or facial* or maxillofacial or jaw) adj3 (injur* or damage* or scars or scarring or scarred or dermatos*)).tw,kf.
34. (keloid* or cicatrix or Acne or Eczema or psoriasis or Epidermolysis Bullosa or port wine stain* or h?emangioma$ or vitiligo or birth mark* or birthmark*).tw,kf.
35. (melanoma* or burns or burn or burned or alopecia or exophthalm* or graves disease or thyroid eye disease or strabismus or (misalign* adj3 eye*) or blepharoptosis or ectropion).tw,kf.
36. ((eye or eyelid or head or neck or mouth or oral) adj3 (neoplasm* or cancer* or tumo?r*)).tw,kf.
37. (Dwarfism or dwarf or small stature or gigantism or restricted growth or torticollis or dystonia or dupuytren* contracture*).tw,kf.
38. (Loss of limb* or amputee* or artificial limb*).tw,kf.
39. (appearance adj3 (abnormal or malformation or problem* or distress)).tw,kf.
40. (visibl* adj3 (differen* or disabilit*)).tw,kf.
41. (deformit* or deformed or disfigured or disfiguring or disfigurement).tw,kf.
42. (palatoschisis or hare lip* or harelip* or (cleft* adj (lip* or palate* or orofacial* or facial or oral or craniofacial))).tw,kf.
43. (Lumpectom* or mastectom* or colostom* or Reconstructive surger* or Reconstruction).tw,kf.
44. or/33-43
45. or/4-32
46. 44 or 45
47. (Camouflage* or make-up or cosmetic* or beauty treatment*).tw,kw. or exp cosmetic/
48. 3 and 46 and 47
49. "randomized controlled trial (topic)"/
50. Randomized Controlled Trial/
51. Randomization/
52. Double Blind Procedure/
53. single blind procedure/
54. placebo/
55. (random* or sham or placebo*).ti,ab,hw.
56. ((singl* or doubl*) adj (blind* or dumm* or mask*)).ti,ab,hw.
57. ((tripl* or trebl*) adj (blind* or dumm* or mask*)).ti,ab,hw.
58. or/49-57

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59. 48 and 58

CINAHL

Search ID#	Search Terms
S66	S53 AND S65
S65	S54 OR S55 OR S56 OR S57 OR S58 OR S59 OR S60 OR S61 OR S62 OR S63 OR S64
S64	TX allocat* random*
S63	(MH "Quantitative Studies")
S62	(MH "Placebos")
S61	TX placebo*
S60	TX random* allocat*
S59	(MH "Random Assignment")
S58	TX randomi* control* trial*
S57	TX ((singl* n1 blind*) or (singl* n1 mask*)) or TX ((doubl* n1 blind*) or (doubl* n1 mask*)) or TX ((tripl* n1 blind*) or (tripl* n1 mask*)) or TX ((trebl* n1 blind*) or (trebl* n1 mask*)
S56	TX clinic* n1 trial*
S55	PT Clinical Trial
S54	(MH "Clinical Trials+")
S53	S43 AND S49 AND S52
S52	S50 OR S51
S51	TI ((Camouflage* or make-up or cosmetic* or beauty treatment*)) OR AB ((Camouflage* or make-up or cosmetic* or beauty treatment*))
S50	(MH "Cosmetics")
S49	S44 OR S45 OR S46 OR S47 OR S48
S48	(MH "Self Concept+")
S47	(MH "Quality of Life+") OR "quality of life"
S46	TI (((Body or self) N2 (confidence or esteem or Image or concept or perception or dysmorphi* or dissatisfaction or shame or anxiety or concern or satisfaction))) OR AB (((Body or self) N2 (confidence or esteem or Image or concept or perception or dysmorphi* or dissatisfaction or shame or anxiety or concern or satisfaction)))

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S45	(MH "Body Dysmorphic Disorder")
S44	(MH "Body Image+")
S43	S1 OR S2 OR S3 OR S4 OR S5 OR S6 OR S7 OR S8 OR S9 OR S10 OR S11 OR S12 OR S13 OR S14 OR S15 OR S16 OR S17 OR S18 OR S19 OR S20 OR S21 OR S22 OR S23 OR S24 OR S25 OR S26 OR S27 OR S28 OR S29 OR S30 OR S31 OR S32 OR S33 OR S34 OR S35 OR S36 OR S37 OR S38 OR S39 OR S40 OR S41 OR S42
S42	TI ((Lumpectom* or mastectom* or colostom* or Reconstructive surger* or Reconstruction)) OR AB ((Lumpectom* or mastectom* or colostom* or Reconstructive surger* or Reconstruction))
S41	TI ((palatoschisis or hare lip* or harelip* or (cleft* N1 (lip* or palate* or orofacial* or facial or oral or craniofacial)))) OR AB ((palatoschisis or hare lip* or harelip* or (cleft* N1 (lip* or palate* or orofacial* or facial or oral or craniofacial))))
S40	TI ((deformit* or deformed or disfigured or disfiguring or disfigurement)) OR AB ((deformit* or deformed or disfigured or disfiguring or disfigurement))
S39	TI ((visibl* N3 (differen* or disabilit*))) OR AB ((visibl* N3 (differen* or disabilit*)))
S38	TI ((appearance N3 (abnormal or malformation or problem* or distress))) OR AB ((appearance N3 (abnormal or malformation or problem* or distress)))
S37	TI ((Loss of limb* or amputee* or artificial limb*)) OR AB ((Loss of limb* or amputee* or artificial limb*))
S36	TI ((Dwarfism or dwarf or small stature or gigantism or restricted growth or torticollis or dystonia or dupuytren* contracture*)) OR AB ((Dwarfism or dwarf or small stature or gigantism or restricted growth or torticollis or dystonia or dupuytren* contracture*))
S35	TI (((eye or eyelid or head or neck or mouth or oral) N3 (neoplasm* or cancer* or tumo?r*))) OR AB (((eye or eyelid or head or neck or mouth or oral) N3 (neoplasm* or cancer* or tumo?r*)))
S34	TI ((melanoma* or burns or burn or burned or alopecia or exophthalm* or graves disease or thyroid eye disease or strabismus or (misalign* N3 eye*) or blepharoptosis or ectropion)) OR AB ((melanoma* or burns or burn or burned or alopecia or exophthalm* or graves disease or thyroid eye disease or strabismus or (misalign* N3 eye*) or blepharoptosis or ectropion))
S33	TI ((keloid* or cicatrix or Acne or Eczema or psoriasis or Epidermolysis Bullosa or port wine stain* or h?emangioma$ or vitiligo or birth mark* or birthmark*)) OR AB ((keloid* or cicatrix or Acne or Eczema or psoriasis or Epidermolysis Bullosa or port wine stain* or h?emangioma$ or vitiligo or birth mark* or birthmark*))
S32	TI (((visible or face or facial* or maxillofacial or jaw) N3 (injur* or damage* or scars or scarring or scarred or dermatos*))) OR AB (((visible or face or facial* or

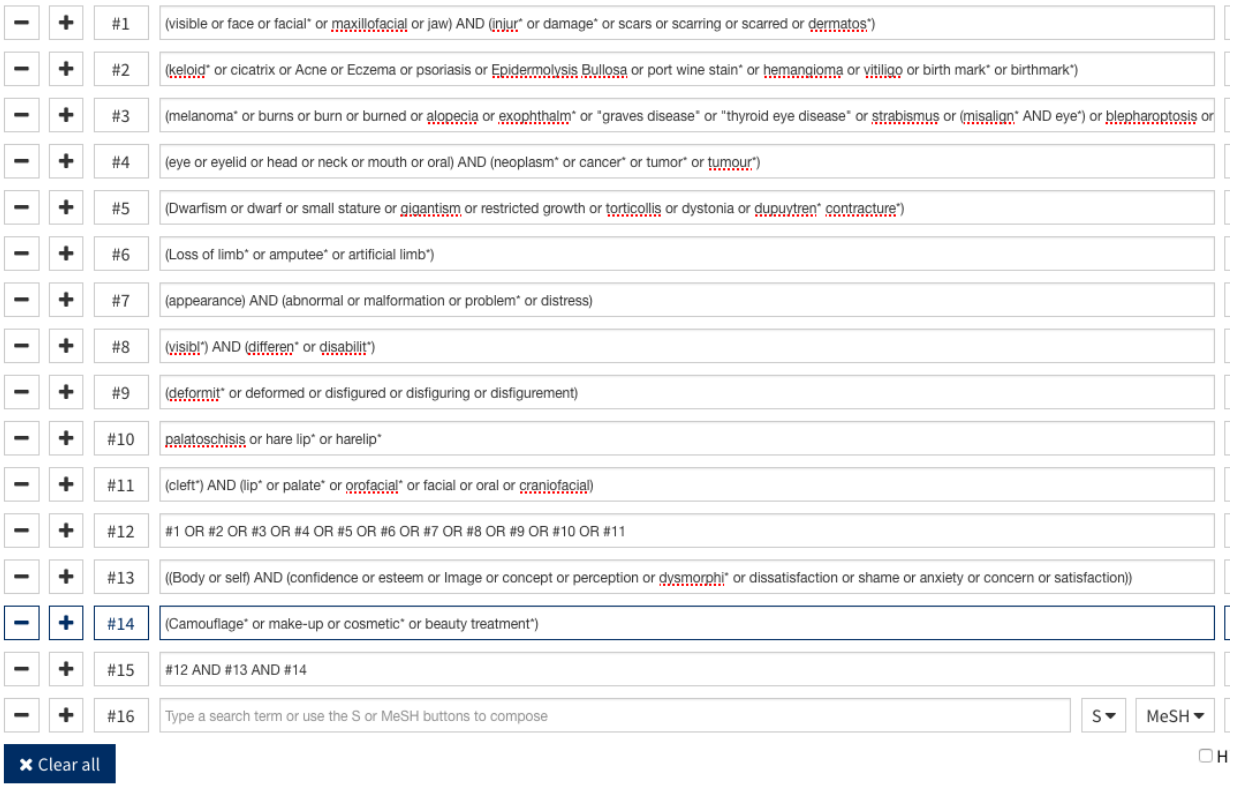
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	maxillofacial or jaw) N3 (injur* or damage* or scars or scarring or scarred or dermatos*)))
S31	(MH "Mouth Abnormalities+")
S30	(MH "Colostomy")
S29	(MH "Mastectomy+")
S28	(MH "Surgery, Reconstructive+")
S27	(MH "Amputees")
S26	(MH "Limb Deformities, Congenital+")
S25	(MH "Limb Prosthesis")
S24	(MH "Amputation+")
S23	(MH "Dupuytren's Contracture")
S22	(MH "Dystonia+")
S21	(MH "Beckwith-Wiedemann Syndrome") OR (MH "Lipodystrophy, Congenital Generalized") OR (MH "Sotos' Syndrome")
S20	(MH "Dwarfism+")
S19	(MH "Growth Disorders+")
S18	(MH "Head and Neck Neoplasms+")
S17	(MH "Mouth Neoplasms+")
S16	(MH "Eyelid Diseases+")
S15	(MH "Strabismus")
S14	(MH "Exophthalmos+")
S13	(MH "Alopecia+")
S12	(MH "Burns") OR (MH "Burns, Chemical") OR (MH "Burns, Electric")
S11	(MH "Melanoma+")
S10	(MH "Pigmentation Disorders+")
S9	(MH "Hemangioma+")
S8	(MH "Skin Abnormalities+")
S7	(MH "Acne Vulgaris")
S6	(MH "Eczema")
S5	(MH "Psoriasis")
S4	(MH "Facial Dermatoses+")

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S3	(MH "Craniofacial Abnormalities+")
S2	(MH "Facial Injuries+")
S1	(MH "Cicatrix+")

Cochrane Central Database of Controlled Trials



#1 (visible or face or facial* or maxillofacial or jaw) AND (injur* or damage* or scars or scarring or scarred or dermatos*)
#2 (keloid* or cicatrix or acne or eczema or psoriasis or Epidermolysis Bullosa or port wine stain* or hemangioma or vitiligo or birth mark* or birthmark*)
#3 (melanoma* or burns or burn or burned or alopecia or exophthalm* or "graves disease" or "thyroid eye disease" or strabismus or (misalign* AND eye*) or blepharoptosis or ectropion)
#4 (eye or eyelid or head or neck or mouth or oral) AND (neoplasm* or cancer* or tumor* or tumour*)
#5 (Dwarfism or dwarf or small stature or gigantism or restricted growth or torticollis or dystonia or dupuytren* contracture*)
#6 (Loss of limb* or amputee* or artificial limb*)
#7 (appearance) AND (abnormal or malformation or problem* or distress)
#8 (visibl*) AND (differen* or disabilit*)
#9 (deformit* or deformed or disfigured or disfiguring or disfigurement)
#10 (palatoschisis or hare lip* or harelip*
#11 (cleft*) AND (lip* or palate* or orofacial* or facial or oral or craniofacial)

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#12 #1 AND #2 AND #3 AND #4 AND #5 AND #6 AND #7 AND #8 AND #9 AND #10 AND #11
#13 (Body or self) AND (confidence or esteem or Image or concept or perception or dysmorphi* or dissatisfaction or shame or anxiety or concern or satisfaction)
#14 (Camouflage* or make-up or cosmetic* or beauty treatment*)
#15 #12 AND #13 AND #14

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APPENDIX 2: **Variables Included in Data Extraction Form**

Study characteristics

First author
Year
Journal
Funding source
Country
Medical condition
Recruitment setting and method
Number of participants randomised
Percent female
Mean (standard deviation [SD]) age
Mean (SD) years with body image concerns
Notes

Intervention and control group details

Group name
Description of intervention
Number of participants randomised to group
Number of participants analyzed in group
Notes

Outcomes

Outcome type (appearance-related, general psychological, reported harms)
Outcome measure name
Timepoints collected
Measure of aggregation (e.g., mean, median) at baseline and subsequent post-test assessments
Measure of dispersion (e.g., SD) at baseline and subsequent post-test assessments
Number of participants analyzed at baseline and subsequent post-test assessments
Other relevant results at baseline and subsequent post-test assessments
Notes

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	3-4
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	6-7
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	7
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	8
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	8
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	8-9
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Supp 1
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	9
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	9-10
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	10, Suppl 2
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	10
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	10-11
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	10-11

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	10
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	NA
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	12
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	28
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	33
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	29-32
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	14-15
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	15-16
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	NA
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	16
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	17-18
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	16-17, 18-19
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	4,26

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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